GRAFFITI^3 : Compact Theory Libraries for Automated Mathematical Discovery

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Abstract

We study *automated conjecturing*: the computer-assisted generation of human-readable symbolic statements intended for expert inspection and subsequent proof or refutation. We formalize conjecture generation as *theory selection* over a finite, evolving evidence base represented by a streaming invariant table, whose rows (objects) and columns (invariants, predicates, and derived features) expand as new constructions and counterexamples are discovered. We present GRAFFITI^3 , a domain-agnostic engine that converts such tables into a *compact theory library*: a budgeted, nonredundant set of snapshot-true bounds, implications, and structural annotations drawn from a controlled grammar. Candidate statements are produced by optimization and then filtered and ranked by explicit simplicity criteria together with dominance-style redundancy elimination, yielding a small library designed for iterative questioning, counterexample search, and downstream deductive work. To support such reasoning, GRAFFITI^3 augments conjectures with short sufficient conditions that explain equality and near-equality regimes, exhaustively verifies each output in the current snapshot, and treats conjectures as provisional until they survive targeted counterexample search or are proved. We also describe a hybrid workflow in which a large language model proposes derived features and narrative summaries while GRAFFITI^3 provides deterministic validation and theory selection. In integer arithmetic, we prove two conjectures produced by this workflow, yielding characterizations of squarefree integers. We demonstrate the methodology on finite groups, knot theory, and a Calabi–Yau hypersurface census, and we revisit graph-theoretic conjectures from the predecessor system *TxGraffiti*, recalling open problems and reporting new ones.

Keywords: Automated conjecturing, Theory formation, Theory selection, *TxGraffiti*, Mathematical invariants, Large language models

"To be a good mathematician, or a good gambler, or good at anything, you must first be a good guesser."

—George Pólya

Contents

1 Introduction **1**

2 Formalizing the Problem of Conjecturing **3**

2.1 A geometric toy example: facets and a conjecture budget 3

2.2 Formal setup: grammars, theories, and budgets 4

2.3 Two failure modes: monster conjectures and encyclopedias 6

2.4 Counterexamples as rolling hold-out tests 6

3 Invariant Design **7**

3.1 LLM Assisted Design 8

4 Graffiti^3 **9**

4.1 From predicates to a hypothesis poset 10

4.2 Quantitative necessary conditions: targets and expression grammar 11

4.3 Heuristic layer: hypotheses, significance, and equality 13

4.4 Sophie layer: sufficient conditions and structural regimes 16

5 Discovery in graph theory **17**

5.1 Established outcomes from the *TxGraffiti* lineage 17

5.2 Re-running the same curated cloud with GRAFFITI^3 18

6 Towards fully automated mathematics **19**

6.1 Integer data: arithmetic separators from snapshots 20

6.2 Finite groups: conjecturing from `SmallGroup` tables 24

6.3 Knots in S^3: invariants and partial geometric regimes 26

6.4 Calabi–Yau threefold hypersurfaces: a theory report artifact 27

7 Conclusion **29**

A Graph-theoretic terms and conventions **30**

B Integer invariants, predicates, and conventions **31**

C Finite-group terms and computational conventions **32**

D Knot-theoretic terms and computational conventions **34**

E Calabi–Yau terms, arithmetic features, and dataset conventions **34**

1 Introduction

We use the term *automated conjecturing* to mean a computational approach to proposing theorem-like mathematical laws from a finite, evolving evidence base. Our goal is not predictive accuracy but a compact library of statements that hold on the current evidence and can be stress-tested by targeted counterexample search. We begin with a structured class of objects in a given domain (for example, graphs, integers, finite groups, or knots) together with numerical invariants and Boolean predicates that describe them. We organize this evidence as a *streaming invariant table*: each row is an object, each column is a descriptor, and both rows and columns grow as new constructions, counterexamples, measurements, and derived features are introduced. Interpreting the numerical columns as coordinates embeds the observed objects as a point cloud in $\mathbb{R}^k$, which we call an *invariant cloud*. The task of a conjecturing system is to organize the invariant cloud with theorem-like statements: bounds that hold on the current table, implications that hold under stated hypotheses, and structural conditions that explain equality and near-equality in recognizable regimes. The goal is not a single function that extrapolates to unseen samples, nor a single "best" conjecture, but a *compact internal theory*: a small, nonredundant library of snapshot-true statements that guides what to study next. In this view, conjectures earn their keep by isolating sharp regimes, naming extremal phenomena, and revealing which hypotheses or invariants are missing when a pattern fails. Automated conjecturing therefore searches for candidate laws *and* the extremal structures that make them sharp, posing questions that a human or automated reasoning system can then try to prove, refute, or explain.

A further methodological point is that the value of a conjecture within an emerging theory does not depend solely on its ultimate truth. Refutations are not dead ends: they help locate the boundary between genuine structure and illusory pattern, and they often suggest how a statement should be strengthened, weakened, or reparameterized. In our setting, the conjecture need not persist for the refutation to be useful: when a counterexample or extremal example is *added as a new row* of the invariant table rather than discarded, it becomes a persistent witness that reshapes subsequent searches. It marks a precise failure mode, alters the feasible region for future conjectures, and can point directly to latent structure (missing hypotheses, alternative invariants, or sharper functional forms). This also changes the role of "leaving data out." Holding out rows may reduce brittleness to accidental artifacts of a snapshot, but it can also withhold the very witnesses that give conjectures their mathematical content, sharpness certificates, near-extremal families, and the first counterexamples that drive repair. Automated conjecturing systems are, therefore, best understood as engines of *theory formation* over streaming invariant tables: they search for candidate laws, identify the instances that make them tight or fail, and iterate as the table and its schema evolve.

This philosophy appears already in early computer systems designed to propose mathematically meaningful statements. An influential example is Siemion Fajtlowicz's *Graffiti* [1], developed in the 1980s at the University of Houston. *Graffiti* generated inequalities between numerical invariants (primarily of graphs) and filtered them using the *Dalmatian* heuristic [2], which demanded both empirical correctness on the current corpus (a *Truth Test*) and mathematical strength (a *Significance Test*). The conjectures disseminated through the *Written on the Wall* series (1980s–1990s) [3–7] seeded substantial work in graph theory and mathematical chemistry and attracted attention from mathematicians including Paul Erdős and Fan Chung. Subsequent systems broadened the methodology and improved accessibility. DeLaViña's *Graffiti.pc* [8] and Larson and Cleemput's *Conjecturing* program [2, 9, 10] extended *Graffiti* with richer grammars and user-friendly interfaces while retaining Dalmatian-style filtering. In parallel, geometric and optimization-driven approaches framed conjectures as extremal constraints in invariant space, including Mélot's *GraPHedron* and *PHOEG* [11] and Hansen and Caporossi's *AutoGraphiX* [12–17]. Together, these systems helped establish automated conjecturing as a practical mechanism for generating and organizing mathematical structure; see [18] for historical context and perspectives.

The present paper is the result of our *Texas Graffiti* (*TxGraffiti*) lineage [19], developed since 2016. *TxGraffiti* paired optimization-driven conjecture generation with heuristic filtering and was deployed through an interactive web application [20]. In graph theory, its conjectures helped seed new theorems in zero forcing, domination, and related topics (e.g., [21–26]). When we applied *TxGraffiti* in domains where we lacked deep subject-matter expertise, its reliance on domain-tailored feature choices and fixed statement forms became a bottleneck. This experience motivated a more flexible, domain-agnostic pipeline that automates a larger share of the conjecturing workflow and reduces dependence on human experts in the loop. Crucially, pursuing portability and deeper automation forced us to formalize what it means to "conjecture well" on a snapshot table: not to return a single strongest statement, but to select a small, nonredundant *theory* under explicit simplicity preferences and a fixed conjecture budget. GRAFFITI^3 implements this redesign and instantiates the resulting theory-selection viewpoint. Inject arbitrary invariant tables, search within a controlled grammar, and select a small, nonredundant conjecture library under explicit simplicity preferences. Beyond quantitative conjectures such as inequalities, it adds a structural layer that accounts for equality and near-equality regimes by identifying simple, interpretable sufficient conditions. We also explore hybrid workflows in which large language models propose derived features, views, and visual summaries with high recall, while GRAFFITI^3 handles mechanical validation and budgeted selection, stabilizing the emerging internal theory.

Contributions. This paper advances automated conjecturing as *theory selection* over streaming invariant tables.

1. We formalize a theory-selection viewpoint in which the output is a *budgeted* library of *snapshot-true* statements drawn from a controlled grammar, emphasizing simplicity and nonredundancy at the level of whole conjecture libraries rather than individual conjectures.

2. We introduce GRAFFITI^3 , a domain-agnostic redesign of the *TxGraffiti* lineage implemented as an open-source Python module within the `txgraffiti` package (available on PyPI [27]), and describe a four-stage pipeline: (a) hypothesis construction from predicates; (b) optimization-driven generation of candidate quantitative statements; (c) heuristic pruning and dominance filtering to enforce a conjecture budget; and (d) a structural layer that searches for simple sufficient conditions explaining equality and near-equality regimes.

3. We demonstrate the architecture across multiple domains—graph invariants, finite groups, integer arithmetic, knot theory, and Calabi-Yau manifolds—showing how budgeted conjecture libraries organize invariant clouds, identify extremal structure, and support hybrid workflows in which large language models propose features and summaries while GRAFFITI^3 provides deterministic validation and selection.

We proceed as follows. Section 2 frames conjecture as a theory-selection problem under incomplete coverage. Section 3 discusses invariant-table design (including missingness) and the role of large language models in feature discovery. Section 4 presents the GRAFFITI^3 pipeline: a controlled grammar, budgeted selection via ranking and dominance filters, and a structural layer that explains equality and near-equality regimes. Sections 5 and 6 report case studies and interpret their results under *snapshot semantics*, emphasizing stress-testing and iterative refinement of conjecture libraries. Finally, Section 7 discusses limitations and future directions, including more autonomous AI-driven discovery loops built around streaming evidence and updated principles of theory.

2 Formalizing the Problem of Conjecturing

Once examples and counterexamples are treated as evolving entries in a common invariant table, a question arises that is not usually posed explicitly in standard supervised learning: *how many* statements that are universal on the current snapshot should a system propose and at what level of generality? In our view, the *number* and *granularity* of reported conjectures play the role of capacity control: the system is not choosing a single predictor to fit a dataset, but choosing a small, readable *theory* to describe the current invariant cloud—the point cloud induced by the numerical columns of the table at a given stage. A central difficulty is that the space of statements consistent with the current table is enormous. Even under a modest grammar, there are typically many inequalities, implications, and structural annotations that are all *universal in the present snapshot* but mutually incomparable in interpretability and explanatory value. Any practical system must therefore solve a *theory selection* problem: it must choose a budgeted, nonredundant subset of conjectures that is informative, legible, and robust to table growth, rather than returning either a single baroque law or an unstructured catalog of true facts.

There is, however, a second difficulty that is easy to miss if one imports the usual machine-learning story too literally. In many mathematical applications the dominant issue is not measurement noise but *coverage*: the target is a fixed (Platonic) mathematical class of objects, typically infinite, and the current table is only a partial and often highly biased view of it. Accordingly, "true on the table" is not a generalization claim; it is a statement about the system's present evidence. Mathematicians manage this gap not by assuming representativeness, but by *stress-testing* candidate laws through active counterexample search, often via constructions engineered to probe extremal structure. This process is not ancillary to conjecturing; it is the mechanism by which the evidence base expands, missing hypotheses are exposed, and patterns are separated from artifacts. An automated conjecturing system faces the same constraint: it must treat conjectures as provisional laws whose value is inseparable from the counterexample search and table growth they provoke.

This suggests an operational interpretation that loosely parallels a modern train/test split but is adapted to the logic of conjecturing. At any moment, write T for the current *snapshot* invariant table. The snapshot T plays the role of a *discovery set* on which candidate laws are proposed and filtered. The analog of evaluation is not a static test set, but a rolling *adversarial audit*: an actively targeted search for objects outside the current table on which proposed laws fail, carried out by construction, optimization, enumeration, or other domain-specific generators. When an audit succeeds, the counterexample is not discarded; it is promoted into the table as a new row, changing the feasible space of subsequent conjectures. In this sense, conjecturing is intrinsically *online*: evidence is refined by an iterative refutation loop rather than by a one-time assumption that an initial sample is representative of an effectively infinite population. We make this rolling-audit viewpoint precise in Subsection 2.4.

2.1 A geometric toy example: facets and a conjecture budget

Before introducing the full grammar-based setup, it is helpful to visualize why automated conjecturing is intrinsically a *theory-selection* problem even in a very simple setting. Suppose a snapshot table records two numerical invariants f_1, f_2 for five objects $o_1, \dots, o_5$, embedding them as points in $\mathbb{R}^2$ as in Figure 1. A basic *linear* grammar of quantitative conjectures consists of halfspace inequalities

$$a\, f_1(o) + b\, f_2(o) \leq c,$$

interpreted as constraints that must hold for every row in the current snapshot.

The convex hull of the resulting invariant cloud makes the core difficulty immediate: there are infinitely many snapshot-true linear inequalities (any halfspace whose boundary line lies above the hull is valid), but only a few are informative, and more specifically, no single inequality is "best". The informative inequalities are those that actively carve out the feasible region, tightening the best-known envelope somewhere. In Figure 1, the red line $\mathcal{H}$ supports a facet (edge) of the hull: it is tight in the snapshot (attained by extremal objects, here o_3 and o_4) and therefore defines one of the strongest linear inequalities witnessed by the current table.

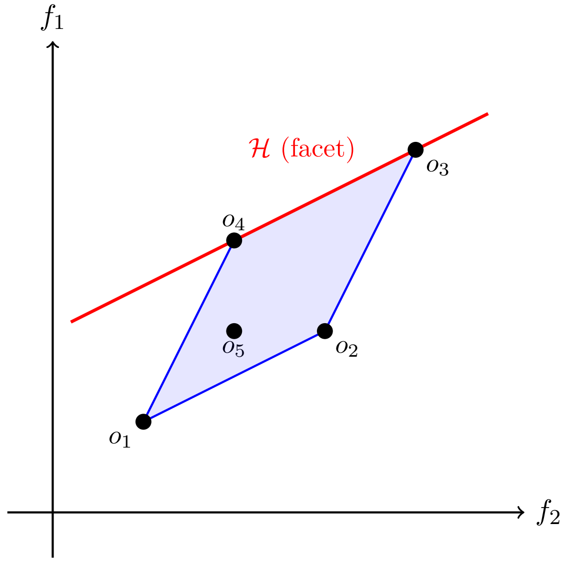

Figure 1: A geometric view of conjectures on a snapshot invariant table. Each point represents an object embedded in invariant space by its numerical invariants. The convex hull defines the feasible region, and its facets correspond to strongest (linear) inequalities supported by the snapshot. The red line $\mathcal{H}$ is one such facet, attained at objects o_3 and o_4. A conjecture budget amounts to keeping only a small set of nonredundant supporting inequalities that each improve the current envelope somewhere, rather than all snapshot-true lines above the hull.

This motivates the role of a *conjecture budget*. Rather than reporting all snapshot-true statements, a system should retain only a small library of nonredundant cuts—for example, at most b facet (or near-facet) inequalities, prioritizing simple ones and discarding candidates that never strengthen the current envelope on any observed object. The remainder of this section abstracts this picture: we replace linear halfspaces by a controlled grammar of admissible formulas and treat the output as a budgeted internal theory that will be re-tested as the table evolves.

2.2 Formal setup: grammars, theories, and budgets

To formalize the theory-selection question, fix a finite collection of objects

$$\mathcal{O} = \{o_1, \dots, o_N\},$$

together with an invariant table T whose rows are indexed by $\mathcal{O}$ and whose columns record numerical invariants and Boolean predicates. The table T should be understood as a *snapshot* of an evolving evidence base: as new objects are constructed or discovered, and as new derived quantities are introduced, both the row set and the column set may grow. Interpreting the numerical columns as coordinates embeds the current objects (where defined) as a point cloud in $\mathbb{R}^k$, which we call the *(current) invariant cloud*. The toy example in Subsection 2.1 is the special case $k = 2$ with conjectures restricted to linear halfspace inequalities.

Let $\mathcal{G}$ denote a formula grammar—a specified collection of syntactic formation rules that determines which expressions and formulas the system is allowed to build. The grammar $\mathcal{G}$ induces a corresponding *language* $\mathcal{L}(\mathcal{G})$ of well-formed conjectures the system is permitted to consider. We restrict attention to conjectures of the form

$$H \Rightarrow \Phi,$$

read as "if H, then Φ." Here H is a Boolean combination of *atomic tests* built from selected columns of T (Boolean predicates and admissible numerical comparisons), and Φ is a Boolean combination of atomic tests built from the *remaining* columns of T, required to hold for every row in the snapshot satisfying H. This enforces a disjointness condition between hypotheses and conclusions: hypotheses describe a regime (a slice of the table), while conclusions describe laws to be tested *within* that regime, rather than definitional rewrites that reuse the same features on both sides. (In the toy geometric example, one may take $H \equiv \text{True}$, so conjectures reduce to unconditional inequalities.) A *theory* is then a finite set

$$C \subseteq \mathcal{L}(\mathcal{G})$$

of conjectures that are snapshot-true on T.

Problem 1 (Fundamental Problem of Automated Conjecturing (theory selection))**.** *Given a snapshot invariant table T over a finite object set $\mathcal{O}$, a formula grammar $\mathcal{G}$ (with induced conjecture language $\mathcal{L}(\mathcal{G})$), and a conjecture budget $b \in \mathbb{N}$, select a theory $C \subseteq \mathcal{L}(\mathcal{G})$ such that*

1. *every $(H \Rightarrow \Phi) \in C$ holds* universally on the snapshot*: for every $o \in \mathcal{O}$, if the row of o in T satisfies H, then it satisfies Φ;*
2. *the budget constraint $|C| \leq b$ holds; and*
3. *among theories satisfying (1)–(2), C is as informative and nonredundant as possible, e.g., it avoids strictly weaker restatements and prioritizes statements that tighten the current feasible region somewhere on the snapshot.*

We do not propose a single canonical objective for Problem 1; instead, we treat it as a guiding design principle. For readers with a machine-learning background, it is helpful to view this as capacity control at the level of *theories* rather than at the level of individual predictors. In this analogy,

- the grammar $\mathcal{G}$ (equivalently, its induced language $\mathcal{L}(\mathcal{G})$) governs the expressivity of admissible conjectures;
- the conjecture budget b plays the role of a theory-size constraint; and
- heuristic filters act as *implicit regularizers*, preferring sharp, nonredundant, and interpretable statements over baroque or locally trivial ones.

This perspective is in the spirit of classical capacity-control viewpoints in statistical learning theory—most notably Vapnik's structural risk-minimization framework [28]—but applied not to a single predictor, rather to a finite library of snapshot-true laws that will be re-tested as the table evolves.

2.3 Two failure modes: monster conjectures and encyclopedias

Two extreme regimes illustrate the tension in Problem 1. At one end, the system may pursue a single conjecture designed to organize the entire snapshot table T; a lone statement of the form $H \Rightarrow \Phi$. If the grammar $\mathcal{G}$ is highly expressive and allows for very complex hypotheses or conclusions, then the system can synthesize an exceedingly intricate Φ (or H) that is true for the snapshot yet effectively encodes idiosyncrasies of the snapshot. The result is a "monster" conjecture: an interpolant of the current evidence disguised as a law. Such statements are brittle: they are tuned so finely to the snapshot that a single new example can shatter them, and their internal structure is too opaque to provide usable mathematical guidance. Worse, in systems that reward formal strength–for example, via Dalmatian-style significance filters (Section 4.3)–monster conjectures can dominate the search, causing simpler, interpretable statements to be discarded as "insignificant" under the heuristic simply because they are weaker.

In the language of machine learning, this is *theory overfitting*. The system has ample expressive capacity but little ability to distribute explanations between multiple statements: with only one conjecture allowed, it absorbs observed regularities by becoming increasingly conditional, baroque, and idiosyncratic to T. Geometrically, a one-statement theory tries to describe a high-dimensional invariant cloud by a single boundary, ignoring the possibility that different regions of the cloud reflect qualitatively different mechanisms and, therefore, call for different laws.

At the other extreme, the system may be permitted to enumerate *every* conjecture in $\mathcal{L}(\mathcal{G})$ that is snapshot-true on the current table T, with no budget and no filtering. The output can easily run to hundreds or thousands of correct statements, many of which are logically redundant, cosmetically variant, or so local that they apply only to tiny subfamilies. Here the failure is opposite in spirit: instead of overcommitting to one universal law, the theory refuses to commit at all. In machine-learning terms, this is *theory underfitting*: the system fails to compress the space of possibilities because each retained conjecture is weak, duplicative, or narrowly scoped, so the collection does not carve the invariant cloud into a legible structure. Mathematically, it resembles a paper that proves many lemmas and special cases without ever extracting a coherent explanation of what is really driving the phenomenon.

2.4 Counterexamples as rolling hold-out tests

The preceding discussion treats the snapshot T as fixed, but the central "generalization" mechanism in conjecturing is that T is not fixed: it grows. This is the mathematically natural resolution of the population-versus-sample tension. A conjecture is not justified by an assumption that the current snapshot is representative; it gains credibility only by surviving an ongoing cycle of attempted refutation that expands the snapshot itself.

One can therefore view the conjecturing loop as maintaining two interacting components. Write T_{disc} for the current *discovery table* (initially, the current snapshot T), on which the system selects a budgeted theory $C \subseteq \mathcal{L}(\mathcal{G})$. In parallel, an *audit process* (a rolling hold-out) searches for objects not already represented in T_{disc} on which conjectures in C fail. If the audit succeeds for some conjecture $(H \Rightarrow \Phi) \in C$, the violating object is a counterexample. Computationally it plays the role of a failed hold-out instance; mathematically it is a refutation. In either reading, the response is the same: the counterexample is promoted into the discovery table as a new row,

$$T_{\text{disc}} \leftarrow T_{\text{disc}} \cup \{\text{counterexample row}\},$$

and the theory is recomputed under the same grammar and budget.

This rolling hold-out viewpoint clarifies why counterexamples are central rather than incidental. A conjecture that is merely snapshot-true is not yet a useful law; it becomes increasingly credible only insofar as it persists under repeated audits that probe regimes not yet well represented in the discovery table: extremal objects, large parameters, sparse or dense limits, highly symmetric constructions, and other adversarially chosen families. Conversely, a conjecture that fails quickly under audit is diagnosed as a locally fitted artifact of incomplete coverage rather than as a stable regularity.

Importantly, the audit process need not be human-driven. It can be instantiated by automated search engines that propose candidate objects: integer-programming and constraint-solving pipelines, enumerators for structured families, reinforcement-learning agents trained with "violate the conjecture" as a reward, or generative models conditioned on target invariant profiles. Following the terminology of [29], we refer to the conjecture-generation component as an *Optimist* and the counterexample-search component as a *Pessimist*, using these terms generically to denote the two roles. Together they yield an end-to-end loop: propose a compact theory, attempt to break it, expand the table, and repeat.

With this addition, the fundamental problem has two coupled axes: (i) how to select a small, nonredundant theory on the current discovery table, and (ii) how to operate a falsification-and growth protocol so that snapshot truth is continually tested against new evidence. The next section discusses one of the main levers available to both axes: *invariant design*. Adding new columns changes the expressive efficiency of the grammar, but it also changes what kinds of audits and counterexample searches are possible, because the system can now target newly defined coordinates when generating or filtering candidate objects.

3 Invariant Design

So far we have treated the snapshot table T and its columns as given, and focused on how many—and what kind of—conjectures a system should return. A second axis of control is the choice of the *invariants themselves*. Adding a new column to T amounts to introducing a new coordinate on the current invariant cloud, often defined as a structured combination of existing columns or as a newly computed descriptor of the underlying object. Well-chosen invariants function like better coordinates: they compress multi-column structure into a single interpretable quantity, so that low-complexity formulas in the grammar $\mathcal{G}$ can express what would otherwise require opaque hypotheses or higher-complexity expressions. In this sense, invariant design governs the *expressive efficiency* of the grammar and directly affects the quality of the budgeted theories $C \subseteq \mathcal{L}(\mathcal{G})$ that a conjecturing engine can discover on a given snapshot.

Invariant design also interacts with the rolling-audit picture of Section 2.4. New columns do not merely enlarge the space of candidate conclusions Φ; they enlarge the space of *audits*. Once a quantity exists as a coordinate of the invariant cloud, it can be used to (i) define new hypotheses H via thresholds, equalities, or discretizations; (ii) define optimization objectives for counterexample search (e.g., "find an object with extreme value of this coordinate that violates the conjecture"); and (iii) target structured families suggested by external systems. Invariant design is therefore a lever both for *compactness* (making important laws expressible in low-complexity form) and for *robustness* (making it easier to probe whether a law is merely snapshot-true or whether it persists under continued table growth).

Historically, much of the success of the *Graffiti* lineage can be read through this lens. Although *Graffiti* itself did not store an explicit table T—it computed invariants on demand during search—its effectiveness relied on a distinctive division of labor. The conjecture forms it explored were intentionally low-complexity (predominantly linear inequalities between invariants), while the numerical *representation* of objects was continually enriched by inventing new descriptors. Indices based on degree sequences, distances, and spectral quantities acted as new coordinates on an implicit invariant cloud, allowing simple admissible formulas to express structure that would otherwise require substantially more complex statements. A concrete example is Fajtlowicz's Havel–Hakimi *residue* $R(G)$: once introduced as a new invariant, *Graffiti* conjectured (correctly) that $R(G)$ is a lower bound for the independence number $\alpha(G)$ (see [30, 31]). In the viewpoint adopted here, such inventions correspond to adding new columns to the (implicit) table, thereby expanding the effective reach of the induced language $\mathcal{L}(\mathcal{G})$.

3.1 LLM Assisted Design

This perspective has direct implications for AI-assisted discovery. Modern large language models, trained on broad mathematical and scientific text, can serve as high-recall *proposal engines* for invariant design: they can suggest new numerical columns and derived Boolean predicates that enrich the coordinate system of the invariant cloud. The conjecturing system then plays the complementary role of *mechanical authority*: it computes the proposed quantities, exhaustively verifies snapshot truth on the current table, and selects a budgeted, nonredundant library under explicit simplicity preferences. The key point is methodological: LLMs are valuable not because they certify truth, but because they broaden the space of candidate coordinates and hypotheses that can then be audited and curated by a conjecturing engine.

We illustrate this concretely with two examples from the *TxGraffiti* interactive web application [20]. In the first, an LLM was prompted to propose a degree-based index built from the 2-*degrees* $d_2(u)$ of a graph, where $d_2(u)$ denotes the number of vertices within distance at most two of u. The proposed quantity was a "reciprocal geometric–arithmetic" edge-sum index on the 2-degree sequence $\{d_2(u)\}_{u\in V(G)}$,

$$RGA^{(2)}(G) = \sum_{uv\in E(G)} \frac{2\sqrt{d_2(u)\,d_2(v)}}{d_2(u)+d_2(v)}.$$

After adding $RGA^{(2)}(G)$ as a new column of T, *TxGraffiti* produced the following open conjecture.

Conjecture 2 (*TxGraffiti*)**.** *If G is a nontrivial connected graph, then $\alpha(G) \le RGA^{(2)}(G)$, and this bound is sharp.*

This example captures the intended division of labor. A human conjecturer would not typically arrive at this particular expression directly as an $\alpha(G)$-bound. However, once the invariant exists as a column of T, the associated inequality is simple to state, exhaustively verify on the current snapshot, and incorporate into a budgeted theory $C \subseteq \mathcal{L}(\mathcal{G})$. More broadly, even when the admissible grammar restricts quantitative conclusions to low-complexity forms in the available columns, expressive derived invariants can make the resulting conjectures mathematically nontrivial.

A second illustration shows a complementary benefit of LLM-assisted feature discovery: once a derived quantity is added as a column of T, it can be used not only in quantitative conclusions but also to *localize* conjectures by defining hypotheses H (thresholds, equalities, and other regime indicators). In a related experiment, after adding 2-degree information, the system discovered a conjecture that holds only under the conjunction of two structural conditions: that G is planar and that $\mathrm{diam}(G) \le 3$. For graphs satisfying this hypothesis, the system proposed an inequality involving the *energy* of a graph,

$$\mathcal{E}(G) := \sum_{i=1}^{n} |\lambda_i(A(G))|,$$

where $\lambda_1(A(G)), \ldots, \lambda_n(A(G))$ are the eigenvalues of the adjacency matrix $A(G)$, and where $\lfloor x \rceil$ denotes rounding x to the nearest integer. The conjectured bound is as follows.

Conjecture 3 (*TxGraffiti*)**.** *If G is a nontrivial connected planar graph with $\operatorname{diam}(G) \leq 3$, then*

$$\lfloor \mathcal{E}(G) \rceil \;\geq\; 3 \sum_{uv \in E(G)} \frac{1}{\sqrt{d_2(u)\, d_2(v)}},$$

and this bound appears sharp on the current evidence base.

This inequality is not universally true on the full table: it fails on at least one graph that does not satisfy the stated hypothesis (see Section 4.3 for the hypothesis-search heuristic used to localize such statements). In our framework, the intended interpretation of Conjecture 3 is therefore as a conditional law of the form $H \Rightarrow \Phi$, where

$$H := \text{“}G \text{ is planar”} \wedge (\operatorname{diam}(G) \leq 3).$$

The example highlights a second role of invariant design: numerical columns not only support quantitative conclusions Φ but also induce useful predicates (e.g., diameter thresholds) that help localize conjectures to regions of the invariant cloud where they persist. Actively searching for such conditions can yield statements that are substantially sharper *within their natural regime* than any unconditional inequality expressible at comparable complexity.

Taken together, these experiments illustrate how large language models can broaden the search space for automated conjecturing by proposing candidate numerical invariants and derived Boolean predicates, thereby enriching both the coordinate system of the invariant cloud and the hypothesis language available to the system. The conjecturing engine can then use this enriched representation to select compact, structured theory libraries C that are easy to miss without systematic feature proposal and mechanical screening.

4 Graffiti^3

At the level of implementation, GRAFFITI^3 is a pipeline that turns a snapshot invariant table into a compact internal theory of quantitative and structural conjectures. The input is a finite object set

$$\mathcal{O} = \{o_1, \ldots, o_N\},$$

together with an invariant table T whose rows are indexed by $\mathcal{O}$ and whose columns record numerical invariants (possibly with missing entries) and optional Boolean predicates. As in the preceding sections, we write the numerical columns as

$$I_1, \ldots, I_k : \mathcal{O} \to \mathbb{R} \cup \{\bot\}$$

and the predicate columns as

$$P_1, \ldots, P_m : \mathcal{O} \to \{\texttt{true}, \texttt{false}\},$$

and we denote their values on o_i by $I_j(o_i)$ and $P_r(o_i)$. The table T is an instance of the snapshot input to Problem 1; the goal is to construct a budgeted, nonredundant theory $C \subseteq \mathcal{L}(\mathcal{G})$ whose conjectures are snapshot-true on T.

The engine proceeds in four conceptual stages:

1. It builds a hypothesis poset from the Boolean columns, viewing a controlled family of Boolean combinations (for example, conjunctions such as "connected and claw-free") as candidate regimes on which to specialize conjectures.

2. For a chosen set of *target* invariants, it solves optimization problems over combinations of the remaining numerical columns (and derived expressions), restricted to rows satisfying each hypothesis, thereby generating a deliberately large pool of snapshot-true conditional inequalities.

3. It ranks and filters these candidates through heuristic criteria (including strength, complexity, and significance tests) to retain a budgeted, nonredundant library in practice, avoiding both overfitting to a single "monster" statement and underfitting via a swarm of redundant inequalities.

4. Finally, it attempts to explain equality and near-equality regimes in structural terms by searching for sufficient conditions: for each surviving bound, it looks for Boolean structure that predicts when the bound is tight (or nearly tight). These sufficient-condition statements are then filtered through a variant of the *Sophie* heuristic introduced by DeLaViña and Waller in [32].

The resulting collection of quantitative bounds and structural conditions is the internal theory associated with the table T and plays the role of a theory $C \subseteq \mathcal{L}(\mathcal{G})$ in Problem 1.

4.1 From predicates to a hypothesis poset

The Boolean columns of the table provide a first vocabulary for describing subclasses of objects. Graffiti[3] begins by identifying predicates that are true on *every* row of the current snapshot. Define

$$\mathcal{P}_{\text{base}} := \{ P_r : P_r(o) = \texttt{true} \text{ for all } o \in \mathcal{O} \}.$$

The *base hypothesis* is the conjunction of these globally true predicates:

$$B := \bigwedge_{P \in \mathcal{P}_{\text{base}}} P,$$

with the convention that $B \equiv \texttt{True}$ if $\mathcal{P}_{\text{base}} = \varnothing$. Predicates in $\mathcal{P}_{\text{base}}$ are not used for further hypothesis specialization and are retained instead as describing the ambient type of the dataset (for example, "connected and nontrivial graphs").

From the remaining Boolean predicates, the system constructs a finite family of hypotheses by taking (conjunctive) refinements of the base hypothesis:

$$h = B \wedge P_{r_1} \wedge \cdots \wedge P_{r_\ell}, \qquad 0 \le \ell \le \ell_{\max},$$

where $r_1, \ldots, r_\ell$ are distinct indices and $\ell_{\max}$ is a small arity bound (typically $\ell_{\max} \in \{1, 2\}$ in practice). Each hypothesis h defines a subclass

$$\mathcal{O}_h = \{ o \in \mathcal{O} : h(o) = \texttt{true} \},$$

and we call $|\mathcal{O}_h|$ the *support* of h (the number of rows of T satisfying h). Two simple pruning rules are applied:

- **Support threshold.** Hypotheses with support smaller than a user-specified (or default) minimum are discarded, preventing conjectures on classes with too little evidence.

- **Redundancy elimination.** If two hypotheses h_1 and h_2 induce the same subclass ($\mathcal{O}_{h_1} = \mathcal{O}_{h_2}$), the longer conjunction (e.g., the one with larger ℓ) is removed.

After pruning, the remaining hypotheses are partially ordered by inclusion of supports:

$$h_1 \preceq h_2 \iff \mathcal{O}_{h_1} \subseteq \mathcal{O}_{h_2},$$

so more specific hypotheses (with smaller support) lie below more general ones. This finite poset is the hypothesis structure on which later stages operate.

Let $\mathcal{H}$ denote the resulting pruned hypothesis family (including the base hypothesis B). The elements of $\mathcal{H}$ are the conditions under which Graffiti^3 will search for quantitative and structural laws. For example, in a graph-invariant table with Boolean columns `connected`, `nontrivial`, `planar`, `regular`, the base hypothesis might be $B = \texttt{connected} \wedge \texttt{nontrivial}$, and $\mathcal{H}$ would include additional hypotheses such as $B \wedge \texttt{planar}$ and $B \wedge \texttt{regular}$, which restrict attention to connected, nontrivial planar graphs and connected, nontrivial regular graphs, respectively[1].

4.2 Quantitative necessary conditions: targets and expression grammar

Once the object set $\mathcal{O}$ and hypothesis family $\mathcal{H}$ are fixed, the engine turns to quantitative statements. Conceptually, Graffiti^3 operates in two stages. First, it deliberately *over-generates* quantitative candidates: for each hypothesis $h \in \mathcal{H}$ and each chosen target invariant, it solves many small optimization subproblems over a controlled family of expression templates in the remaining numerical columns. This produces a high-recall pool of short inequalities intended to act as quantitative *necessary* conditions, i.e., statements that are snapshot-true on the restricted row set $\mathcal{O}_h$ (equivalently, conditionals of the form $h \Rightarrow \Phi$).

Second, heuristic filters (the *Hazel*, *Morgan*, and *Dalmatian* layers, defined in Section 4.3) rank, prune, and deduplicate this raw collection to obtain a small library of bounds that are both simple and meaningfully sharp somewhere in the table. The *Sophie* layer operates *after* this pruning: it treats tightness and near tightness events for the surviving bounds as new targets when searching for structural *sufficient* conditions.

A *target* is a designated numerical invariant

$$I^\star : \mathcal{O} \to \mathbb{R}$$

that appears on the left-hand side of the quantitative conjectures, while the remaining numerical invariants provide building blocks for right-hand side expressions. For each hypothesis $h \in \mathcal{H}$, these statements are generated and verified on the corresponding slice $\mathcal{O}_h$.

In general, Graffiti^3 can process multiple targets sequentially and then apply the ranking and pruning heuristics to the union of the resulting candidate sets; for clarity, we restrict attention here to a single target. When the user is unsure what to target, Graffiti^3 can optionally propose candidates via a separate selection module (e.g., based on variance, extremal behavior, or external criteria such as the `GAMBIT` subsystem [33]); it can also query an LLM for suggestions (see Section 6). In all cases, the current target $I^\star$ is removed from the pool of available invariants when constructing right-hand side expressions.

For a fixed hypothesis $h \in \mathcal{H}$, write

$$\mathcal{O}_h = \{\, o \in \mathcal{O} : h(o) = \texttt{true} \,\}$$

[1]This is the first instance where Graffiti^3 differs from *TxGraffiti*. In the previous system conjunctions on Boolean columns were always hard-coded prior to conjecturing.

for the slice of objects on which h holds (equivalently, the set of rows of T satisfying h). Quantitative conjectures are expressed in the canonical conditional form

$$\forall o \in \mathcal{O} \quad h(o) \Rightarrow I^\star(o) \ \bowtie \ f(\mathbf{I}^{\neg\star}(o)), \qquad \bowtie \in \{\leq, \geq, =\}, \tag{1}$$

where $\mathbf{I}^{\neg\star}$ denotes the vector of numerical invariants other than $I^\star$, and f is drawn from a controlled family of expression templates in these remaining invariants. Equivalently, (1) asserts that $I^\star(o) \bowtie f(\mathbf{I}^{\neg\star}(o))$ for all $o \in \mathcal{O}_h$ (on rows where the required invariants are defined). Right-hand sides in (1) are not fixed by hand. Instead, GRAFFITI[3] organizes their construction into a collection of *expression stages*. Each stage specifies:

- a small feature map
$$\Psi(o) = (\phi_1(o), \ldots, \phi_d(o)),$$
where each ϕ_j is a simple transformation of the non-target invariant vector $\mathbf{I}^{\neg\star}(o)$ (for example a coordinate $I_a(o)$, $\sqrt{I_b(o)}$, $\log I_b(o)$ when defined, or two-coordinate combinations such as $\sqrt{I_a(o)I_b(o)}$ or $\log(I_a(o) + I_b(o))$); and
- a procedure that, for each hypothesis h and target $I^\star$, searches for affine bounds of the form
$$I^\star(o) \ \bowtie \ w_1\phi_1(o) + \cdots + w_d\phi_d(o) + b \quad \text{for all } o \in \mathcal{O}_h,$$
subject to box constraints on the coefficients w_j and intercept b[2].

In all stages the feature dimension d is kept small (typically $d \in \{1, 2\}$) so that (i) the resulting expressions are short, and (ii) the underlying subproblems are inexpensive enough to solve repeatedly across many hypotheses and feature choices. When a feature transformation is not defined on some rows (e.g., log), we restrict attention to the sub-slice where all features used by the stage are defined.

Linear optimization stages. Most stages share the same linear-programming template. For a fixed h and target $I^\star$, view the rows in $\mathcal{O}_h$ as pairs (x_i, y_i) with $x_i = \Psi(o_i) \in \mathbb{R}^d$ and $y_i = I^\star(o_i) \in \mathbb{R}$. An *upper bound* of the form $I^\star(o) \leq w^\top \Psi(o) + b$ is obtained by solving

$$\min_{w,b,s} \ \sum_i s_i \quad \text{subject to} \quad w^\top x_i + b - s_i = y_i, \ \ s_i \geq 0, \ \ |w_j|, |b| \leq M, \tag{2}$$

where M is a stage-specific coefficient bound. The constraints imply $w^\top x_i + b \geq y_i$ for all i, and the ℓ_1 objective pushes the affine function toward the upper envelope of the point cloud $\{(x_i, y_i)\}$ while respecting the box constraints. A symmetric formulation with $w^\top x_i + b + s_i = y_i$ produces *lower bounds* $I^\star(o) \geq w^\top \Psi(o) + b$.

After solving, real coefficients are rationalized (replaced by nearby rationals with small denominators) and filtered for "human niceness" (bounded magnitudes and denominators). The surviving affine combination $w^\top \Psi + b$ is then converted back into an explicit expression f in the original invariants. In a separate verification step, every candidate inequality is checked on all rows of $\mathcal{O}_h$, and any expression that fails on even a single row is discarded. Thus, before heuristic pruning, the quantitative layer produces only inequalities that are empirically universal on the current table slice.

[2]Taken together, this is the second instance where GRAFFITI[3] differs from *TxGraffiti*. In the previous system feature engineering columns were always hard-coded prior to conjecturing.

A convex-hull stage. In addition to the LP-based stages, GRAFFITI[3] includes an optional convex-hull stage that incorporates the facet-selection viewpoint of *GraPHedron* and *PHOEG* into the same pipeline. For a small subset of numerical invariants $\{X_1, \dots, X_\ell\}$ that includes the target $I^\star$, the system considers the point cloud

$$\{(X_1(o), \dots, X_\ell(o)) : o \in \mathcal{O}_h\}$$

(optionally after simple rescaling of coordinates) and computes its convex hull. Facet-defining inequalities

$$a_1 X_1 + \cdots + a_\ell X_\ell \le c$$

with nonzero coefficient on $I^\star$ are then rewritten so that $I^\star$ appears on the left-hand side, yielding conjectures of the form $I^\star \le f(X_2, \dots, X_\ell)$ or $I^\star \ge f(X_2, \dots, X_\ell)$. The resulting coefficient vectors are normalized and rationalized, and candidates failing basic "human niceness" filters are discarded. In low dimensions this recovers the classical selection principle of *GraPHedron* and *PHOEG*, but here it is one stage among many and can be combined with the LP-based stages.

Optional pairwise derived invariants. When enabled, GRAFFITI[3] augments the table with a small family of pairwise derived numerical columns. For each unordered pair (X, Y) of numerical invariants that is *empirically noncomparable on the current table*—meaning that there exist rows $o, o' \in \mathcal{O}$ with $X(o) < Y(o)$ and $X(o') > Y(o')$—the system constructs

$$|X - Y|, \qquad \min(X, Y), \qquad \max(X, Y), \qquad X \cdot Y,$$

and makes these quantities available to expression stages whose templates permit them (restricted to rows where the operations are defined).

This augmentation is most informative precisely in the noncomparable case: because the sign of $X - Y$ changes across rows, $\min(X, Y)$ and $\max(X, Y)$ isolate regimes where one invariant dominates the other, while $|X - Y|$ highlights near-equality structure and $X \cdot Y$ captures joint scale effects. From the optimization viewpoint, these are additional coordinates that can be included in feature maps $\Psi(o)$; from the conjecturing viewpoint, they expand the space of short, recognizable templates to include expressions such as

$$I^\star \le a \min(X, Y) + b\sqrt{|X - Y|} + c \quad \text{or} \quad I^\star \ge a\, X \cdot Y + b \max(X, Y) + c,$$

which become natural once the corresponding quantities exist as columns of the invariant table.

4.3 Heuristic layer: hypotheses, significance, and equality

By the time conjectures reach the heuristic layer, snapshot correctness has already been enforced: for a fixed hypothesis h, any candidate relation that fails on any row of the slice $\mathcal{O}_h$ (on which the expression is defined) is discarded. The survivors are treated as *raw conjecture candidates* and annotated with metadata, including:

- the support size $|\mathcal{O}_h|$ of the hypothesis slice;
- a *touch number*, i.e., the count of rows in $\mathcal{O}_h$ on which the relation is attained to within a fixed numerical tolerance (applied consistently across a run); and
- basic statistics of the pointwise slack (e.g., minimum and median slack on $\mathcal{O}_h$), with slack defined in the natural direction for $\le$ or $\ge$ bounds.

Even for a single (target, hypothesis) slice, the quantitative stage can generate hundreds or thousands of correct inequalities that largely overlap. The purpose of the heuristic layer is to compress this raw pool into a small, legible set without collapsing to a brittle "best" statement.

Operationally, this layer combines two filters and one ranking mechanism:

1. the *Morgan heuristic*, which generalizes hypotheses by merging duplicate relations that arise on multiple subclasses (after canonicalization of expressions);

2. a Dalmatian-style significance filter, which discards inequalities that are redundant because they never improve the current envelope on $\mathcal{O}_h$; and

3. a bookkeeping and ordering rule (*Hazel*) that ranks survivors using equality information, primarily by touch number.

Together, Morgan and Dalmatian act as theory-level regularizers: they enforce a practical conjecture budget by suppressing duplicate hypotheses and locally dominated inequalities, while Hazel provides a stable notion of priority for reporting and for the subsequent structural layer.

Morgan: generalizing hypotheses. The Morgan filter operates at the level of hypotheses rather than numerical strength. Write a quantitative conjecture in canonical form as

$$C: \; h \Rightarrow R,$$

where $h \in \mathcal{H}$ is a Boolean hypothesis and R is a fixed quantitative relation (including target, direction, and a canonicalized right-hand side), e.g.,

$$R \equiv \big(I^\star \le f(\mathbf{I}^{\neg\star})\big) \quad \text{or} \quad R \equiv \big(I^\star \ge f(\mathbf{I}^{\neg\star})\big).$$

Morgan groups conjectures by their relation R (ignoring the hypothesis) and compares hypotheses by the subsets of rows on which they hold. Concretely, each h induces the index set

$$M_h \;=\; \{\, i \in \{1, \dots, N\} : h(o_i) = \texttt{true} \,\} \qquad (\text{equivalently, } \mathcal{O}_h = \{o_i : i \in M_h\}).$$

Within a fixed relation R, suppose we have conjectures

$$h_1 \Rightarrow R, \quad h_2 \Rightarrow R, \quad \dots, \quad h_s \Rightarrow R,$$

with masks $M_{h_1}, \dots, M_{h_s}$. Morgan keeps only those hypotheses whose support is maximal by inclusion:

$$h_i \text{ is kept if there is no } j \text{ such that } M_{h_i} \subsetneq M_{h_j}.$$

Intuitively, if the same quantitative relation R holds on a strictly larger slice of the current table, then the more specialized versions do not increase empirical coverage on this snapshot and are removed. For example, if an inequality discovered under "connected and triangle-free" also holds under "connected," Morgan prefers the more general statement and discards the specialization.

This behavior targets the "too many hypotheses" failure mode: without Morgan pruning, the theory accumulates copies of the same relation annotated by nested Boolean descriptions. With Morgan pruning, each relation is retained only on hypotheses whose support cannot be expanded further on the current table.

Dalmatian: local significance within a hypothesis. After Morgan pruning, the remaining conjectures can still be numerous. The Dalmatian filter then operates *within* each triple $(I^\star, \bowtie, h)$

of target, direction, and hypothesis to enforce a local significance criterion. Fix $h \in \mathcal{H}$ and write $\mathcal{O}_h = \{o \in \mathcal{O} : h(o) = \texttt{true}\}$. Consider candidate upper bounds for a fixed target $I^\star$ on $\mathcal{O}_h$,

$$I^\star(o) \le u_1(o),\ I^\star(o) \le u_2(o),\ \ldots, \qquad o \in \mathcal{O}_h,$$

generated by the quantitative stages. Dalmatian processes candidates in a fixed deterministic order (e.g., increasing expression complexity, then by tightness on the table), maintaining a running envelope u_* defined pointwise by

$$u_*(o) = \min\{u_j(o) : j \in A\},$$

where A is the set of indices accepted so far. A new candidate u is accepted if and only if it is *strictly better somewhere* beyond a fixed tolerance $\varepsilon > 0$:

$$\exists\, o \in \mathcal{O}_h \ \text{ such that } \ u(o) < u_*(o) - \varepsilon.$$

Lower bounds are handled symmetrically by maintaining a running maximum envelope and requiring strict improvement on at least one object. Candidates that never improve the current envelope are discarded, even though they are universally correct on $\mathcal{O}_h$. In the language of Section 2, Dalmatian prevents the theory from hoarding inequalities that do not further restrict the invariant cloud on the current table slice: every surviving bound sharpens the best-known envelope on at least one example.

Equality slices, touch numbers, and Hazel ordering. Equality and near-equality events play a special role in how Graffiti[3] prioritizes quantitative conjectures and in how it prepares inputs for the structural (sufficient-condition) layer. Consider an accepted upper bound

$$C: \quad h(o) \Rightarrow I^\star(o) \le f(\mathbf{I}^{\neg\star}(o)),$$

valid on the slice $\mathcal{O}_h$. The system records the *equality slice*

$$\mathsf{Eq}(C) = \Big\{\, o \in \mathcal{O}_h : |I^\star(o) - f(\mathbf{I}^{\neg\star}(o))| \le \varepsilon_{\mathrm{eq}}(o) \,\Big\},$$

and its *touch number* $\mathrm{touch}(C) = |\mathsf{Eq}(C)|$, where $\varepsilon_{\mathrm{eq}}(o)$ is a numerical tolerance (chosen consistently across targets, e.g., via a fixed absolute tolerance after normalization, or a mixed absolute/relative tolerance). It also records a complementary *strict slice*

$$\mathsf{Str}(C) = \Big\{\, o \in \mathcal{O}_h : I^\star(o) < f(\mathbf{I}^{\neg\star}(o)) - \varepsilon_{\mathrm{eq}}(o) \,\Big\},$$

which captures objects where the inequality is valid but not near-tight under h (for lower bounds the strict inequality is reversed). Note that $\mathsf{Eq}(C)$ and $\mathsf{Str}(C)$ are disjoint; the remaining rows in $\mathcal{O}_h$ fall into a tolerance band between near equality and strict slack.

Hazel is then a bookkeeping rule that orders conjectures using equality information and other user-chosen priorities. In the current implementation, Hazel sorts conjectures primarily by decreasing touch number (optionally normalized by support $|\mathcal{O}_h|$) and uses secondary keys such as support size and slack statistics to break ties. More generally, Hazel accepts an arbitrary scoring functional, so conjectures can be surfaced according to any desired notion of "interestingness." In particular, unreleased versions integrate additional ranking signals from the Inequality Inference and Ranking System (`IRIS`) [34]. Hazel is an ordering heuristic rather than a correctness or significance test: it does not change which conjectures survive Morgan and Dalmatian, but it controls the presentation order and the priority with which equality slices are forwarded to the structural layer.

4.4 Sophie layer: sufficient conditions and structural regimes

At the end of the heuristic layer, the surviving quantitative conjectures are still interpreted as *necessary conditions*: inequalities and equalities that are universally valid on the relevant table slice(s). Morgan and Dalmatian reduce the raw candidate pool to a small, nonredundant library, and Hazel imposes a stable priority order using touch numbers and related statistics. The *Sophie* module then takes over. Its purpose is to attach *sufficient* structural explanations to high-priority numerical regimes of these bounds—most prominently equality (tightness) and strictness (non-tightness) on the current table—thereby augmenting the quantitative theory with interpretable Boolean annotations. The design is inspired by the Sophie heuristic of DeLaViña and Waller [32].

All Sophie statements are formed inside the *base universe* determined by the base hypothesis B. Write

$$\mathcal{O}_B \;=\; \{\, o \in \mathcal{O} : B(o) = \texttt{true} \,\}.$$

Sophie draws on two inputs restricted to $\mathcal{O}_B$: (i) the Boolean columns of T, which provide candidate structural descriptors, and (ii) the quantitative conjectures that survived Morgan and Dalmatian together with the equality/strict slices they induce. The output consists of short implications of the form "if this structural pattern holds, then this numerical regime occurs," which we interpret as *sufficient-condition annotations* for the quantitative library.

Sophie works with two Boolean vocabularies on $\mathcal{O}_B$. *Structural hypotheses* H are built from the Boolean columns of T restricted to $\mathcal{O}_B$. After excluding predicates that are trivial on $\mathcal{O}_B$ (always true or always false) and excluding predicates already absorbed into the base hypothesis B, Sophie forms a candidate family using a low-arity closure, typically:

- single properties Q (e.g., `bipartite`);
- their negations $\neg Q$ (e.g., `non-bipartite`);
- small conjunctions $Q_1 \wedge Q_2$ (optionally up to arity $\le \ell_{\max}$), subject to a minimum support threshold.

Each hypothesis H is represented by its mask $M_H = \{\, o \in \mathcal{O}_B : H(o) = \texttt{true} \,\}$; if distinct formulas induce the same mask on $\mathcal{O}_B$, Sophie keeps only the syntactically simpler one.

Event targets P are derived from the surviving quantitative conjectures. Each quantitative conjecture can be written schematically as

$$C: \quad h_C(o) \Rightarrow L_C(o) \;\bowtie\; R_C(o), \qquad \bowtie \in \{\le, \ge, =\},$$

where $h_C \in \mathcal{H}$ is the hypothesis under which C was generated and L_C, R_C are numerical expressions in the invariants. Sophie converts C into Boolean event predicates on $\mathcal{O}_B$ by restricting attention to the *active slice* $\mathcal{O}_B \cap \mathcal{O}_{h_C}$ and then forming regime tests (up to numerical tolerances), such as

$$L_C = R_C, \quad L_C < R_C, \quad L_C > R_C, \quad L_C \le R_C, \quad L_C \ge R_C,$$

as well as near-equality via $|L_C - R_C| \le \delta$ when enabled. These event masks are the numerical regimes Sophie aims to *explain* structurally.

Given a structural hypothesis H and an event target P, Sophie tests the empirical implication

$$H \Rightarrow P \qquad \text{on } \mathcal{O}_B,$$

equivalently mask inclusion $M_H \subseteq M_P$ within the base universe (up to tolerances and optional violation allowances). For each pair (H, P) it records:

- the support of H, $\mathrm{supp}(H) = |M_H|$;
- the coverage of P by H, $\mathrm{cov}(H, P) = |M_H \cap M_P|$; and
- the number of violations, $\mathrm{viol}(H, P) = |M_H \setminus M_P|$.

Only pairs meeting user-specified thresholds (minimum support for H and for the target P, minimum coverage, and a maximum allowed violation count—often zero) are retained as candidate Sophie conditions. When the reverse implication $P \Rightarrow H$ also holds on the current table, Sophie records the resulting biconditional $H \iff P$ as a candidate characterization of the numerical regime.

5 Discovery in graph theory

This section describes an ongoing program in automated conjecturing for graph theory built around a long-evolving, human-curated invariant table. Here, the "research loop" is documented through the resulting sequence of publications: conjectures generated from the current invariant cloud are then proved, refuted, or refined, and the examples and counterexamples arising in that process are incorporated as new rows (and occasionally new columns) in the table, shaping subsequent conjecturing runs. The aim is therefore not to freeze a single snapshot and report its output, but to show how conjecture generation and mathematical follow-through co-evolve over time.

We use the *TxGraffiti* lineage (2016–2023) to summarize representative results from earlier phases of this program, and then present several conjectures produced by GRAFFITI^3 that were out of reach of the earlier workflow (for example, because GRAFFITI^3 supports a more flexible hypothesis pipeline and a systematic structural layer). Throughout, G denotes a finite, simple, connected graph on at least two vertices. Definitions of graph invariants and predicates used here appear in Appendix A.

5.1 Established outcomes from the *TxGraffiti* lineage

The invariant cloud underlying this section was curated over years through a standard research loop: targeted searches for extremal graphs, counterexamples discovered during proof attempts, and iterative inclusion of families suggested by the literature. The *TxGraffiti* system was originally designed to mine such clouds for low-complexity inequalities that appear stable under repeated table expansion and adversarial counterexample search. Several of its highest-value outputs have since been proved and published.

On the slice of connected r-regular graphs with $r > 0$, *TxGraffiti* isolated the inequality $\alpha(G) \leq \mu(G)$, linking the independence number and the matching number on a natural structural regime. What made this output striking at the time (2017) is that $\alpha(G)$ and $\mu(G)$ are among the oldest and most intensively studied extremal parameters in graph theory, and regular graphs form one of the most classical testbeds; nevertheless, this clean inequality does not seem to have been recorded in the literature in this generality. Prompted by the regular-case discovery, the conjecture was proved and strengthened to the following general bound, which subsumes the regular case as an immediate corollary.

Theorem 4 (Caro, Davila, and Pepper [35]; *TxGraffiti*)**.** *If G is a connected and nontrivial graph, then $\delta(G)\,\alpha(G) \;\leq\; \Delta(G)\,\mu(G)$, and this bound is sharp. In particular, if G is r-regular with $r > 0$, then $\alpha(G) \leq \mu(G)$.*

A second class of notable outputs concerns sharp upper bounds for zero forcing and its variants in bounded-degree regimes. In the cubic setting, *TxGraffiti* produced conjectures that ultimately required long, structured proofs, reflecting the combinatorial complexity of forcing processes even in degree-3 graphs.

Theorem 5 (Davila and Henning [23]; *TxGraffiti*)**.** *If $G \neq K_4$ is a connected cubic graph, then $Z(G) \leq 2\gamma(G)$, and the bound is sharp.*

Theorem 6 (Davila [25]; *TxGraffiti*)**.** *If G is a connected, cubic, claw-free graph, then $Z(G) \leq \gamma(G) + 2$, and the bound is sharp.*

Theorem 7 (Davila and Henning [22]; *TxGraffiti*)**.** *If $G \neq K_{3,3}$ is a connected cubic graph, then $Z_t(G) \leq \frac{3}{2}\gamma_t(G)$, and the bound is sharp.*

These conjectures illustrate a recurring pattern in our loop: a snapshot-clean inequality can hide substantial extremal structure. In several instances, the resulting proofs are long (often approaching twenty pages) and hinge on a careful analysis of equality and near-equality regimes.

Beyond these results, additional conjectures from the *TxGraffiti* lineage have been proved and published, including results linking independence to domination-type parameters in regular or forbidden-subgraph classes, and results connecting zero forcing to vertex cover and related parameters on structured families. Rather than provide a catalogue here, we point the reader to [24–26] (and references therein), which document extremal families, algorithmic consequences, and comparisons with previous bounds.

Not all stable-looking snapshot laws have been resolved. A notable open problem generated by this workflow concerns the relationship between zero forcing and independence on subcubic graphs:

Conjecture 8 (*TxGraffiti* and Graffiti^3)**.** *If $G \neq K_4$ is a connected graph with maximum degree $\Delta(G) \leq 3$, then $Z(G) \leq \alpha(G) + 1$, and the bound is sharp.*

Conjecture 8 has accumulated substantial partial progress, including confirmation on important subclasses and verification on broad computational slices of the cubic regime; see, for example, [21, 36, 37]. Within the logic of automated conjecturing, it functions as a long-running stress test: it guides which graph families are injected into the table and where counterexample search is concentrated.

5.2 Re-running the same curated cloud with Graffiti^3

The redesigned Graffiti^3 engine retains the classical *TxGraffiti*-style quantitative workflow—searching for snapshot-true bounds on hypothesis slices—but adds ingredients:

- automatic detection of Boolean hypothesis and non-redundant conjunctions, and
- automatic feature engineering of nonlinear conjectures, and
- a budgeted selection layer that enforces a compact, nonredundant library under explicit simplicity and dominance-style nonredundancy criteria; and
- a structural ("Sophie") layer that turns numerically distinguished *events* (tightness and near-tightness slices of strong bounds, sharp thresholds, or other designated regime masks) into explicit sufficient-condition statements.

When run on the same invariant cloud and target choices used throughout the *TxGraffiti* lineage, Graffiti^3 re-discovers many of the stable inequalities that historically organized the cloud (for example, relationships among α, μ, β, domination parameters, and Z described above). This provides an end-to-end calibration: the newer engine recovers the main quantitative laws that survived years of counterexample search, table growth, and subsequent proof attempts—in several cases leading to theorems with substantial, multi-page arguments.

In addition, the Sophie layer produces outputs of a different type. Rather than only reporting bounds, it extracts compact structural annotations that explain high-priority numerical regimes observed on the cloud. These statements should be read with snapshot semantics: they are exhaustively correct on the current table, but their purpose is to compress a visible pattern into a clean conjecture that can be directly stress-tested. For example, the following snapshot-true conjectured graph planarity condition in-terms of its zero forcing number.

Conjecture 9 (Graffiti^3)**.** *If G is a nontrivial and connected graph with $Z(G) < 4$, then G is planar.*

The graph-theoretic thread differs from the later case studies in one key respect: the invariant cloud here is heavily curated. Its rows and families were accumulated over years through extremal search, counterexample discovery during proof attempts, and deliberate injection of structured constructions from the literature. In the next section we move to non-graph domains where the invariant tables are less curated and the available hypotheses may be thinner. The goal there is not to claim the same level of long-horizon stability, but to show how the same pipeline behaves under snapshot semantics in new settings: how it produces compact, auditable conjecture libraries, what kinds of failures appear first, and how stress-testing can drive the next round of table refinement or proof discovery.

6 Towards fully automated mathematics

The graph-theoretic case study of Section 5 rests on an invariant table whose rows were curated over years through domain expertise: targeted searches for extremal objects, counterexamples discovered during proof attempts, and sustained interaction with the literature. The case studies in this section deliberately remove that advantage. Our aim is not to present polished results in mature domains, but to document a reproducible workflow for forming compact internal theories from snapshot tables assembled quickly, with limited manual curation and without assuming deep specialist intuition. These experiments are not yet fully autonomous: the author acts as an explicit orchestrator, while the conjecture search, verification, and budgeted selection are fully automated once the table schema and target are fixed.

All examples below follow the same discovery loop, with the author mediating between a large language model (LLM) and Graffiti^3 ; see Figure 2.

1. *Domain, dataset, and schema proposal (LLM $\to$ author).* The author prompts an LLM to propose (a) a tractable domain and a finite object universe, (b) a concrete snapshot-generation plan (including code), and (c) a schema consisting of computable invariants and predicates. The author also fixes a reference for terminology (either a standard textbook definition set, or an appendix of explicit definitions).

2. *Snapshot construction (author).* The author executes (and, when necessary, minimally edits) the code to build a deterministic snapshot invariant table T with one row per object.

3. *Targeting and conjecture generation* (Graffiti^3). The LLM specifies a target invariant(s) $I^\star$ and provides $(T, I^\star)$ to Graffiti^3 , which over-generates candidate inequalities and Sophie-style structural statements, verifies them exhaustively on the snapshot, and selects a small, nonredundant library under an explicit complexity budget.

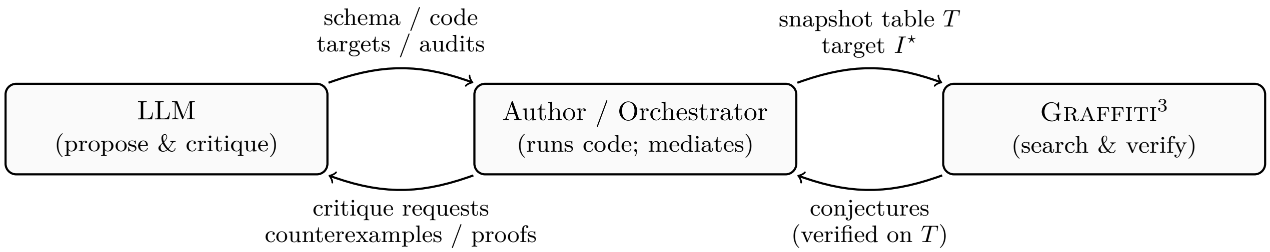

Figure 2: Orchestrated discovery loop shared by the case studies in Section 6. The author mediates all communication: the LLM proposes representations and audit strategies; GRAFFITI^3 returns snapshot-verified conjectures; the LLM helps triage, search for counterexamples, and suggest proof routes.

4. *Mathematical triage and adversarial audit (LLM ↔ author).* The author passes the resulting conjecture library back to the LLM for critique: redundancy detection, plausibility assessment, proposed proof mechanisms, and targeted counterexample searches via cutoff increases and structured family injections. The author executes these audits, updates or extends the snapshot table, and reruns the pipeline as needed. Statements that survive repeated audits become plausible research targets; others are retired once explicit counterexamples are found.

Across the case studies, GRAFFITI^3 plays the same role: given a finite snapshot table T, it proposes quantitative inequalities and Sophie-style statements that are *snapshot-universal* under their stated hypotheses (no violations on any active row of T). We therefore present outputs in the standard mathematical register: as conjectures supported by exhaustive computation on the current snapshot. The audit step is part of the story: some snapshot-verified statements survive adversarial holdouts and become plausible research targets, while others are quickly falsified once natural counterexample families are injected. When a statement admits a short independent proof, we promote it to a theorem and include the proof explicitly.

6.1 Integer data: arithmetic separators from snapshots

To remove the "curated table" advantage present in the graph case study, we construct a snapshot table from a simple, rapidly generated dataset of integers. The author prompted an LLM to propose (i) a finite integer universe large enough to exhibit nontrivial multiplicative structure, (ii) a menu of standard arithmetic invariants and Boolean predicates computable from prime factorizations, and (iii) code to populate a table with these quantities. We then fixed the universe and schema once, and computed all table entries deterministically. Concretely, the object universe is

$$\mathcal{O} \;=\; \{\,2, 3, \ldots, 2 \cdot 10^6\,\}.$$

Let T denote the resulting *snapshot invariant table*, with one row per $n \in \mathcal{O}$.

All arithmetic functions used in this subsection are standard; for background and conventions see, e.g., Apostol's *Introduction to Analytic Number Theory* [38] or Hardy–Wright's *An Introduction to the Theory of Numbers* [39]. For completeness and reproducibility we also record the exact definitions and predicate conventions in Appendix B.

For each $n \in \mathcal{O}$ we record the numerical invariants

$$\varphi(n),\ \lambda(n),\ \mathrm{rad}(n),\ \omega(n),\ \Omega(n),\ \tau(n),\ \sigma(n),$$

where φ is Euler's totient, λ is the Carmichael function, $\mathrm{rad}(n)$ is the radical, $\omega(n)$ and $\Omega(n)$ count prime factors without/with multiplicity, $\tau(n)$ is the divisor-counting function, and $\sigma(n)$ is the divisor-sum function. We also record Boolean predicates including

$$\texttt{is\_prime}(n),\ \texttt{is\_squarefree}(n),\ \texttt{is\_almost\_squarefree}(n),\ \texttt{is\_7\_smooth}(n),$$

where `is_almost_squarefree`(n) abbreviates $\Omega(n) - \omega(n) = 1$ and `is_7_smooth`(n) means all prime factors of n are at most 7.

Finally, we append a controlled collection of simple derived columns (deterministic combinations of the invariants), e.g. the totient defect $n - \varphi(n)$, the ratio $\mathrm{rad}(n)/n$, and basic normalizations of $\sigma(n)$ and $\tau(n)$. These derived coordinates enlarge the candidate search space available to GRAFFITI^3 while keeping the entire snapshot schema explicit.

Following the LLM's recommendation to use a canonical arithmetic target with rich structure, we set the target invariant to

$$I^\star(n) = \varphi(n),$$

with base hypothesis $B \equiv (n \geq 2)$. We then ran GRAFFITI^3 on $(T, I^\star)$ under a fixed complexity budget. The engine over-generates candidate inequalities and Sophie-style structural statements, verifies them exhaustively on the snapshot, and then returns a small nonredundant library. In particular, it returned (among other items) a bounded list of low-complexity statements that act as *separators*: inequalities whose truth on the snapshot is equivalent to a Boolean predicate (notably squarefreeness).

We relayed the GRAFFITI^3 library back to the LLM in an explicit adversarial role: propose redundancy eliminations, identify likely known facts, suggest proof strategies, and (crucially) search for counterexamples by increasing cutoffs and injecting structured families (e.g. prime powers, smooth numbers, and highly composite patterns). This audit immediately recovered standard relations such as

$$\varphi(n) \geq \lambda(n), \qquad \varphi(n) \leq n - 1,$$

as well as snapshot biconditionals like

$$\big(\varphi(n) = n - 1\big) \iff \texttt{is\_prime}(n),$$

which serve as an end-to-end sanity check: the pipeline rediscovers classical statements from a generic invariant table without injecting number-theoretic rules.

In addition to the above mentioned inequalities, the Sophie layer flagged related equality events holding throughout the tested range, including

$$\big(\varphi(n) = n - 1\big) \iff \texttt{is\_prime}(n), \qquad \big(\varphi(n) = \mathrm{bnd}(n)\big) \iff \texttt{is\_prime}(n),$$

suggesting that primes lie exactly on the surface $\varphi(n) = \mathrm{bnd}(n)$ in $(n, \mathrm{rad}(n), \varphi(n))$-space.

Among the most informative outputs were two snapshot-verified separators that partition the point cloud

$$\big\{(n, \mathrm{rad}(n), \varphi(n)) : n \in \mathcal{O}\big\}$$

into squarefree and nonsquarefree regimes using inequalities of very low algebraic complexity. Define

$$\mathrm{bnd}(n) \ = \ 4\sqrt{n\,\mathrm{rad}(n)} - (3n + 1), \qquad \Delta(n) \ = \ \mathrm{bnd}(n) - \varphi(n).$$

On the entire snapshot universe $\mathcal{O}$, GRAFFITI^3 returned the following biconditionals with no violations:

$$\varphi(n) \ \leq \ \mathrm{bnd}(n) \quad \iff \quad \texttt{is\_squarefree}(n), \tag{3}$$

$$\varphi(n) \ \leq \ n - \mathrm{rad}(n) \quad \iff \quad \neg\texttt{is\_squarefree}(n). \tag{4}$$

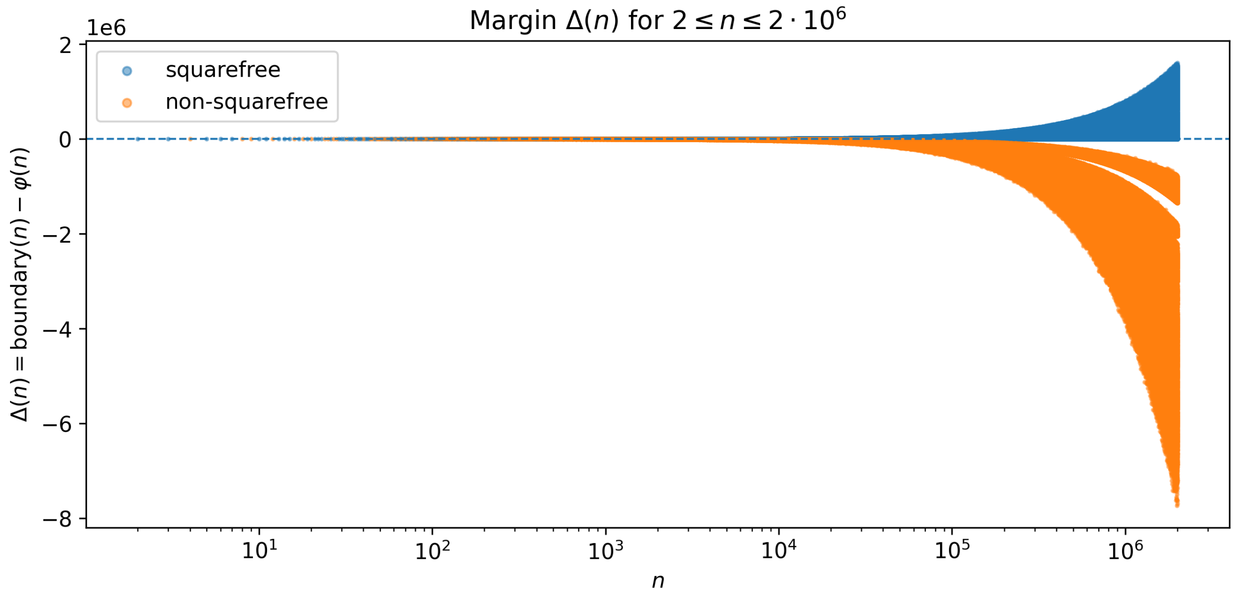

Figure 3: Empirical behaviour of $\Delta(n) = \mathrm{bnd}(n) - \varphi(n)$ for $2 \le n \le 2 \cdot 10^6$. On this tested range, the sign of $\Delta(n)$ separates squarefree from nonsquarefree integers with no violations, consistent with the snapshot-verified biconditional $\Delta(n) \ge 0 \iff \texttt{is\_squarefree}(n)$.

Equivalently, (3) is

$$\Delta(n) \ge 0 \quad \iff \quad \texttt{is\_squarefree}(n),$$

so the sign of $\Delta(n)$ yields a single scalar "event predicate" reusable in subsequent conjecture generation.

In addition, the Sophie layer flagged equality events holding throughout the tested range, including

$$(\varphi(n) = n - 1) \iff \texttt{is\_prime}(n), \qquad (\varphi(n) = \mathrm{bnd}(n)) \iff \texttt{is\_prime}(n),$$

suggesting that primes lie exactly on the surface $\varphi(n) = \mathrm{bnd}(n)$ in $(n, \mathrm{rad}(n), \varphi(n))$-space.

The LLM adversarial audit (see Figure 2) flagged (3) and (4) as "open-looking" separators: they are not standard textbook inequalities in this explicit form, yet survived targeted counterexample searches. In this instance, both separators admit short direct proofs. The linear separator (4) was proved by the author after the snapshot run. The curved separator (3) was proved by D. Romik (personal communication) after inspecting the conjecture. We therefore upgrade these two snapshot-verified statements to proved results.

Theorem 10 (Romik; Graffiti[3]). *If $n \in \mathbb{Z}$ with $n \ge 2$, then n is squarefree if and only if*

$$\varphi(n) \ \le\ 4\sqrt{n\ \mathrm{rad}(n)} - (3n + 1).$$

Moreover, this inequality holds with equality if and only if n is prime.

Proof. Assume first that n is squarefree. Then $\mathrm{rad}(n) = n$, hence

$$4\sqrt{n\ \mathrm{rad}(n)} - (3n + 1) = 4\sqrt{n^2} - (3n + 1) = 4n - (3n + 1) = n - 1,$$

where $\sqrt{n^2} = n$ since $n > 0$. Thus the inequality becomes $\varphi(n) \le n - 1$, which holds for all $n \ge 2$. Moreover, equality holds if and only if $\varphi(n) = n - 1$, which happens if and only if n is prime.

Conversely, suppose that n is not squarefree. Then there exists a prime p with $p^2 \mid n$. Write $n = p^a m$ with $a \ge 2$ and $\gcd(p, m) = 1$. By multiplicativity of rad on coprime factors,

$$\mathrm{rad}(n) = \mathrm{rad}(p^a)\,\mathrm{rad}(m) = p\;\mathrm{rad}(m).$$

Since $\mathrm{rad}(m) \le m$, we have $\mathrm{rad}(n) \le pm$. Also $a \ge 2$ implies $n = p^a m \ge p^2 m$, hence $\frac{n}{p} \ge pm$, so

$$\mathrm{rad}(n) \le pm \le \frac{n}{p} \le \frac{n}{2}.$$

Therefore,

$$4\sqrt{n\;\mathrm{rad}(n)} - (3n+1) \le 4\sqrt{n \cdot \frac{n}{2}} - (3n+1) = (2\sqrt{2} - 3)n - 1 < 0.$$

But $\varphi(n) \ge 1$ for all $n \ge 2$, so the inequality $\varphi(n) \le 4\sqrt{n\;\mathrm{rad}(n)} - (3n+1)$ cannot hold when n is not squarefree (and equality is impossible in this case). $\square$

Theorem 11 (Davila; Graffiti[3]). *If $n \in \mathbb{Z}$ with $n \ge 2$, then n is nonsquarefree if and only if*

$$\varphi(n) \;\le\; n - \mathrm{rad}(n).$$

Moreover, equality $\varphi(n) = n - \mathrm{rad}(n)$ holds if and only if $n = p^2$ for a prime p.

Proof. First suppose that n is squarefree. Then $\mathrm{rad}(n) = n$, hence $n - \mathrm{rad}(n) = 0$, while $\varphi(n) \ge 1$ for every $n \ge 2$. Thus $\varphi(n) \le n - \mathrm{rad}(n)$ fails.

Conversely, assume that n is not squarefree. Write

$$n = \prod_{i=1}^{t} p_i^{a_i},$$

where the p_i are distinct primes and $a_j \ge 2$ for at least one index j. Then

$$\mathrm{rad}(n) = \prod_{i=1}^{t} p_i \qquad \text{and} \qquad \varphi(n) = n \prod_{i=1}^{t} \Big(1 - \frac{1}{p_i}\Big).$$

Define

$$m := \prod_{i=1}^{t} p_i^{a_i - 1},$$

so that $n = \mathrm{rad}(n)\, m$ and $m \ge 2$. Using the totient formula,

$$\varphi(n) = \mathrm{rad}(n)\, m \prod_{i=1}^{t} \Big(1 - \frac{1}{p_i}\Big) = m\, \varphi(\mathrm{rad}(n)).$$

Also,

$$n - \mathrm{rad}(n) = \mathrm{rad}(n)(m - 1).$$

Hence $\varphi(n) \le n - \mathrm{rad}(n)$ is equivalent (after dividing by $\mathrm{rad}(n) > 0$) to

$$m \prod_{i=1}^{t} \Big(1 - \frac{1}{p_i}\Big) \le m - 1.$$

Choose an index j with $a_j \geq 2$. Then p_j divides m, so $m \geq p_j$. Moreover, since all factors lie in $(0, 1)$, we have

$$\prod_{i=1}^{t}\Big(1 - \frac{1}{p_i}\Big) \leq 1 - \frac{1}{p_j}.$$

Multiplying by m and using $m \geq p_j$ gives

$$m\prod_{i=1}^{t}\Big(1 - \frac{1}{p_i}\Big) \leq m\Big(1 - \frac{1}{p_j}\Big) \leq m\Big(1 - \frac{1}{m}\Big) = m - 1,$$

which proves $\varphi(n) \leq n - \mathrm{rad}(n)$ for all nonsquarefree n.

For the equality statement, suppose $\varphi(n) = n - \mathrm{rad}(n)$. Tracing the inequalities above, equality forces

$$\prod_{i=1}^{t}\Big(1 - \frac{1}{p_i}\Big) = 1 - \frac{1}{p_j} \quad \text{and} \quad m = p_j.$$

If $t \geq 2$, then the product is strictly smaller than each individual factor, so necessarily $t = 1$. Thus, $n = p^a$ for a prime p, and then $m = p^{a-1} = p$ forces $a = 2$, hence $n = p^2$. Conversely, if $n = p^2$, then $\varphi(n) = p^2 - p$ and $\mathrm{rad}(n) = p$, so $\varphi(n) = n - \mathrm{rad}(n)$ holds. □

Not all GRAFFITI^3 snapshot-true conjectures were resolved in our experiments. We therefore record here a small set of representative open conjectures. Among the unresolved statements, the LLM flagged the following as highest priority for further investigation.

Conjecture 12 (GRAFFITI^3)**.** *If $\varphi(n) > 4\,\lambda(n)$, then n is not perfect.*

Conjecture 13 (GRAFFITI^3)**.** *Fix a base $b \geq 2$ (empirically $b = 2$). If $\varphi(n) \leq \frac{9}{19}\,n$, then n is not a Fermat pseudoprime to base b.*

Conjecture 14 (GRAFFITI^3)**.** *For all $n \geq 1$,*

$$\varphi(n) \geq 2\tau(n) - 8.$$

We now turn from integer arithmetic to the finite-group experiments.

6.2 Finite groups: conjecturing from `SmallGroup` tables

Throughout this subsection, G denotes a finite group. To build a group table, we use the `SmallGroup` identifiers provided by the `SmallGrp` package in `GAP` [40–42]. Fix an order cutoff N (in our runs $N = 128$), and let

$$\mathcal{O} \;=\; \{\, G_{n,k} : 1 \leq n \leq N,\ 1 \leq k \leq \#\texttt{SmallGroup}(n) \,\}, \qquad G_{n,k} = \texttt{SmallGroup}(n, k).$$

Let T denote the corresponding *snapshot invariant table*, with one row per $G \in \mathcal{O}$. For each $G \in \mathcal{O}$ we record the numerical invariants

$$k(G),\ |Z(G)|,\ |G'|,\ |G/G'|,\ |G|,\ \exp(G),$$

where $k(G)$ is the number of conjugacy classes, $Z(G)$ is the center, G' is the derived subgroup, G/G' is the abelianization, and $\exp(G)$ is the exponent. We also include Boolean predicates

$$\texttt{is\_abelian}(G),\ \texttt{is\_cyclic}(G),\ \texttt{is\_nilpotent}(G),\ \texttt{is\_solvable}(G),\ \texttt{is\_pgroup}(G).$$

We take $I^\star(G) = k(G)$ as the target invariant. On the snapshot table T we ran GRAFFITI^3 with base hypothesis $B \equiv (|G| \geq 1)$. Among the highest-ranked snapshot-universal outputs are standard inequalities such as

$$k(G) \geq |Z(G)|, \qquad k(G) \geq |G/G'|, \qquad k(G) \leq |G|.$$

These serve as end-to-end calibration: from a generic invariant table, the pipeline recovers canonical relationships among conjugacy classes, the center, abelianization, and order without injecting group-theoretic rules by hand.

We then subjected the selected library to an explicit adversarial stress test. In particular, an LLM (GPT-5.2) was instructed to act in an adversarial role: propose redundancy eliminations, flag likely rediscoveries, and attempt falsification by (a) enlarging the order cutoff N when feasible and (b) injecting structured holdout families outside the initial `SmallGroup` window, with an emphasis on non-solvable examples (e.g., alternating groups and groups of Lie type, when available via external constructions). A representative stable output flagged during the stress test were two seperate conjectures involving a square-root boundary built from $|Z(G)|$ and $|G|$. These conjectures are combined in the following statement.

Conjecture 15 (GRAFFITI^3)**.** *The following two statements are conjectured to hold for all finite groups G:*

(a) ***Separator clause (solvability):***

$$k(G) \geq \frac{1}{4}\,|Z(G)| + \frac{3}{4}\sqrt{|Z(G)|\,|G|} \quad \Longrightarrow \quad G \text{ is solvable.}$$

(b) ***Envelope clause (nilpotent region):***

$$G \text{ nilpotent} \quad \Longrightarrow \quad k(G) \geq \frac{1}{4}\,|Z(G)| + \frac{3}{4}\sqrt{|Z(G)|\,|G|}.$$

Moreover, on the snapshot universe $\mathcal{O}$ the envelope clause holds and the inequality $k(G) \geq \frac{1}{4}\,|Z(G)| + \frac{3}{4}\sqrt{|Z(G)|\,|G|}$ is sharp (i.e., there exist rows attaining equality).

Conjecture 15 is naturally framed in terms of the *commuting probability*

$$\mathrm{cp}(G) \;:=\; \Pr_{x,y\in G}(xy = yx),$$

which is linked to conjugacy classes by the identity

$$\mathrm{cp}(G) \;=\; \frac{k(G)}{|G|}.$$

See, e.g., [43, 44] for background and sharp threshold results. Because the center contributes commuting pairs in a structurally unavoidable way, it is also natural to introduce the *center proportion*

$$z(G) \;:=\; \frac{|Z(G)|}{|G|}.$$

Dividing the inequality in Conjecture 15 by $|G|$ yields an equivalent criterion in the two-dimensional invariant space $(z(G), \mathrm{cp}(G))$.

Conjecture 16 (GRAFFITI^3)**.** *If* $z(G) := |Z(G)|/|G|$ *and*

$$\mathrm{cp}(G) \ \geq \ \tfrac{1}{4}z + \tfrac{3}{4}\sqrt{z}, \tag{5}$$

then G *is solvable.*

In the same `SmallGroup` runs, the system proposed additional clean-looking inequalities relating $k(G)$ to commutator structure (e.g., via $\sqrt{|G'|}$ with affine correction terms). Several were snapshot-universal on $\mathcal{O}$ but were quickly falsified once standard non-solvable families were injected. We omit these unstable artifacts and record only Conjecture 15 as a representative statement that remained stable under our adversarial stress tests. For other open standing snapshot-true conjectures, see Appendix C.

6.3 Knots in S^3: invariants and partial geometric regimes

Throughout this subsection, $K \subset S^3$ denotes a knot and $u(K)$ denotes its unknotting number. The author prompted an LLM to propose (i) a tractable knot dataset (a census at a specified cutoff) and (ii) a menu of standard invariants and predicates that can be extracted mechanically from existing databases and software. Once $\mathcal{K}$ and the schema were fixed, all table entries were populated deterministically from the chosen sources and scripts (with missing values recorded explicitly rather than imputed).

Let T denote the resulting *snapshot invariant table*, with one row per $K \in \mathcal{K}$. Unlike the integer and finite-group case studies, some columns are populated only when auxiliary geometric computations succeed; we therefore treat T as a *current evidence snapshot* whose coverage can improve over time. All knot-theoretic quantities used below are standard; for background and conventions see, e.g., Rolfsen's *Knots and Links* [45] or Lickorish's *An Introduction to Knot Theory* [46]. For reproducibility we also record the exact term conventions and missing-value logic used in this subsection in Appendix D. For each $K \in \mathcal{K}$ we record numerical invariants

$$c(K), \quad \det(K), \quad \sigma(K), \quad v_2(K), \quad v_3(K),$$

and, when available, the hyperbolic volume $Vol(K)$.

We also record Boolean predicates

$$\texttt{is\_hyperbolic}(K), \qquad \texttt{fibered}(K),$$

and a *data-availability* predicate

$$\texttt{has\_Vol}(K),$$

meaning that the table entry for $Vol(K)$ is present (not missing/NaN) on the row for K. This predicate is an engineering object: it tracks coverage of the geometric pipeline rather than a topological property. The target invariant (suggested by the LLM (GPT-5.2)) for conjecturing is $u(K)$.

We asked GRAFFITI^3 to produce a budgeted library of snapshot-verified lower bounds for $u(K)$, together with Sophie-style implications intended to act as *regime hypotheses* on the current snapshot (in particular, predicates involving `fibered` and `has_Vol`). As elsewhere, the engine over-generates candidate inequalities valid on T and then applies the heuristic layer (Section 4.3) to retain a small, nonredundant set. The highest-ranked universal inequality was the signature obstruction

$$u(K) \ \geq \ \frac{1}{2}\,|\sigma(K)|,$$

serving as an end-to-end calibration rather than a novelty claim (see Appendix D for the signature convention used, and standard references for the obstruction). The raw library also contained stronger formulas that involve only $|\sigma(K)|$ under partial-regime hypotheses such as $\texttt{has\_Vol}(K)$ and $\texttt{fibered}(K) \wedge \texttt{has\_Vol}(K)$, including expressions growing quadratically in $|\sigma|$. Before recording such statements, we subjected them to an adversarial audit.

Here, the LLM suggested standard families and scaling heuristics likely to generate counterexamples once $\mathcal{K}$ is enlarged. In particular, bounds of the form $u \geq c|\sigma|^2 + O(|\sigma|)$ were flagged as growth-unstable: along many natural families, both $u(K)$ and $|\sigma(K)|$ are expected to grow at most linearly in diagrammatic complexity, so a quadratic-in-$|\sigma|$ lower bound should be expected to fail once the regime contains knots with moderately large signature. We therefore treat such formulas as snapshot-shape artifacts and do not promote them as durable conjectures.

After discarding ill-typed and growth-unstable candidates, the strongest remaining output in this run is a signature-based envelope inside the $\texttt{fibered} \cap \texttt{has\_Vol}$ slice. Since $u(K) \in \mathbb{Z}$, we record the inequality in explicitly integer-typed form:

Conjecture 17 (GRAFFITI^3: a fibered, volume-available signature envelope)**.** *On the current snapshot, if* $\texttt{fibered}(K)$ *and* $\texttt{has\_Vol}(K)$ *then*

$$u(K) \;\geq\; \left\lceil \frac{3}{4}\,|\sigma(K)| \right\rceil \;-\; \left\lceil \frac{18}{17}\sqrt{|\sigma(K)|} \right\rceil \;+\; 1.$$

Conjecture 17 is intentionally presented as a *snapshot-verified regime envelope* rather than as a universal statement about all knots. Its methodological value is to define a precise audit target: within the portion of the current table where (i) the knot is labeled fibered and (ii) a volume entry is available, the snapshot suggests that $u(K)$ is controlled more rigidly by signature than the universal obstruction $u(K) \geq \frac{1}{2}|\sigma(K)|$.

6.4 Calabi–Yau threefold hypersurfaces: a theory report artifact

Throughout this subsection, $X_{\mathbf{w}} \subset \mathbb{P}^4(\mathbf{w})$ denotes a (quasi-smooth) Calabi–Yau threefold hypersurface specified by a weight vector $\mathbf{w} = (w_1, \ldots, w_5)$. We use a standard census of weighted-projective Calabi–Yau threefold hypersurfaces. The author prompted an LLM to propose (i) a canonical object universe drawn from an existing list, (ii) a small menu of low-complexity arithmetic features of $\mathbf{w}$ together with topological outputs recorded in the census, and (iii) a deterministic extraction script that assembles these quantities into a snapshot table. Once the dataset source and schema were fixed, all table entries were computed deterministically. As elsewhere, we treat the resulting table T as a *current evidence snapshot*: statements returned by GRAFFITI^3 are verified on T under their stated hypotheses, but are not asserted a priori as theorems beyond the dataset.

Let $\mathbf{w} = (w_1, \ldots, w_5)$ be a weight vector (sorted nondecreasingly $w_1 \leq \cdots \leq w_5$) and set the anticanonical degree

$$d \;:=\; \sum_{i=1}^{5} w_i.$$

Each hypersurface carries Hodge numbers $(h^{1,1}, h^{2,1})$, and we define the Euler characteristic

$$\chi \;:=\; 2(h^{1,1} - h^{2,1}).$$

For each $X_{\mathbf{w}}$ in the census we record numerical columns

$$(w_1, \ldots, w_5), \quad d, \quad h^{1,1}, \quad h^{2,1}, \quad \chi.$$

We also record Boolean predicates capturing coarse arithmetic types of $\mathbf{w}$:

- `SD`($\mathbf{w}$) (*sum divisible*): $d \equiv 0 \pmod{w_i}$ for all i.
- `EO`($\mathbf{w}$) (*exactly one even*): exactly one w_i is even.
- `REP`($\mathbf{w}$) (*has repeat*): the multiset $\{w_1, \ldots, w_5\}$ has a repeated value.
- `PC`($\mathbf{w}$) (*pairwise coprime*): $\gcd(w_i, w_j) = 1$ for all $i < j$.

These predicates are deterministic functions of $\mathbf{w}$, are extremely low-complexity, and are intended to serve as reusable regime hypotheses. Background on quasi-smooth weighted-projective Calabi–Yau hypersurfaces and the canonical datasets appears in the Kreuzer–Skarke / PALP literature and related resources; see Appendix E for a detailed description of definitions included in this case study.

We asked GRAFFITI^3 to generate budgeted conjecture libraries with targets among χ and $h^{1,1}$ on the snapshot table T. We relayed the top GRAFFITI^3 outputs to an LLM (GPT-5.2) in an explicit adversarial role: remove definitional rewrites (e.g. inequalities that simply restate $\chi = 2(h^{1,1} - h^{2,1})$), flag likely-vacuous consequences of dataset-wide constraints, and propose stress tests and plausible counterexample families. After filtering out tautologies and obviously unstable artifacts, we retained a small collection of snapshot-verified statements that were both high-ranked by GRAFFITI^3 and recommended by the adversarial LLM as worth recording.

Conjecture 18 (GRAFFITI^3)**.** *For any $X_{\mathbf{w}} \subset \mathbb{P}^4(\mathbf{w})$ Calabi–Yau threefold hypersurface*

1. (Divisibility $\Rightarrow$ nonpositive Euler characteristic.) *If `SD`($\mathbf{w}$) holds, then $\chi \leq 0$.*
2. (Exactly-one-even $\Rightarrow$ a sharp linear $h^{1,1}$ threshold.) *If `EO`($\mathbf{w}$) holds, then*
$$h^{1,1} \;\geq\; \frac{d}{6} - 1,$$
equivalently $\chi \geq -2h^{2,1} + \frac{1}{3}d - 2$.
3. (Threshold exclusion I.) *If $h^{1,1} \leq \frac{d}{6} - 1$, then `PC`($\mathbf{w}$) fails.*
4. (Threshold exclusion II.) *If $h^{1,1} < \frac{d}{6} - 1$, then `EO`($\mathbf{w}$) fails.*
5. (Repeat-weight slope bound.) *If `EO`($\mathbf{w}$) and `REP`($\mathbf{w}$) hold, then*
$$14\,h^{1,1} \;\geq\; 13\,w_3 + 1,$$
where w_3 is the median weight under $w_1 \leq \cdots \leq w_5$.

For this dataset we took one extra step beyond the standard loop. After GRAFFITI^3 generated the conjecture library and the LLM completed adversarial screening, we prompted the LLM to *synthesize* the retained conjectures into a compact internal "theory" for the run: to organize the statements into themes, propose a unifying explanatory mechanism (where appropriate), and write a self-contained LaTeX research-note-style document that (i) states the conjectures, (ii) explains why they are not mere definitional rewrites, and (iii) proposes concrete falsification experiments. The author/orchestrator then executed the resulting LaTeX code to produce the PDF included in this paper's companion GitHub repository.[3]

[3] `https://github.com/RandyRDavila/TxGraffiti2/tree/main/papers/graffiti3`

7 Conclusion

Automated conjecturing systems have repeatedly shown the same pattern: simple machine-generated statements can catalyze genuine progress once they are interpreted, sharpened, and stress-tested. GRAFFITI^3 continues this lineage, but makes a structural commitment explicit: it is a *theory learner*. Rather than optimizing a single predictor, it enforces capacity at the level of an entire conjecture library, returning a finite, readable internal theory: a budgeted set of bounds, implications, and structural annotations that are mechanically verified on the current evidence snapshot.

The case studies in this manuscript are intentionally methodological. They demonstrate that the same end-to-end loop can operate across integer arithmetic, finite groups, graph invariants, knots, and Calabi–Yau data without rewriting domain-specific logic. What matters is not any one conjecture in isolation, but the inquiry-selected theory that remains after simplicity, dominance, significance, and Sophie-style regime filters are applied. This is the core separation we advocate: generation can be expansive, but selection must be conservative, auditable, and budgeted. In this separation, the system supports machine-generated discovery while retaining a stable interface for human interpretation and proof.

In our framework, counterexamples function as adversarial hold-out tests: when a conjecture fails, the violating object is promoted into the table, reshaping subsequent searches. Looking ahead, the main direction is to reduce the amount of human orchestration in this loop. The experiments here are not yet fully autonomous: the author mediates between a high-recall proposal engine (for features, targets, and narrative critique) and GRAFFITI^3 as the deterministic validator and selector. A next step is to strengthen the "pessimist" side of the pipeline by providing more sophisticated methods of counterexample search. In parallel, richer but still auditable invariant design, improved handling of missingness, and better theory-level ranking signals can further stabilize what is retained under a fixed conjecture budget. Taken together, these directions aim for a more autonomous discovery loop built around streaming evidence: propose, select, attempt to break, expand the table, and repeat.

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Statements and Declarations

Competing Interests

The author declares that there are no competing interests.

Funding

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Author Contributions

The author conceived the study, designed and implemented the experiments, analyzed the results, and wrote the manuscript.

Data Availability

All data generated or analyzed during this study are included in this published article and its supplementary information files.

Code Availability

Code used to generate the experiments and figures is available at the GitHub repository for [27].

Ethics Approval

Not applicable.

Consent to Participate

Not applicable.

Consent for Publication

Not applicable.

A Graph-theoretic terms and conventions

Throughout Section 5, all graphs are finite, simple, and undirected. For a graph G, we write $V(G)$ and $E(G)$ for its vertex set and edge set, respectively. The *order* of G is $n(G) = |V(G)|$, and the *size* of G is $m(G) = |E(G)|$. Two vertices $v, w \in V(G)$ are *neighbors, or* adjacent, if $v, w \in E(G)$. The *degree* of a vertex $v \in V(G)$ is the number of neighbors of v. The minimum and maximum degrees of G are denoted by $\delta(G)$ and $\Delta(G)$. Two edges are said to be *incident* if they share a common endpoint (vertex).

The complete graph with order n is denoted K_n. The star graph with $n-1$ leaves is denoted $K_{1,n-1}$. The claw graph is $K_{1,3}$. A graph is said to be H*-free* if it does not contain H as an *induced* subgraph. A graph G is *connected* if every pair of vertices is joined by a path. A graph G is r*-regular* if every vertex has degree r. The case $r = 3$ is called *cubic*. A graph is *subcubic* if $\Delta(G) \leq 3$. A graph G is *bipartite* if $V(G)$ can be partitioned into two independent sets X and Y (called *parts*) such that every edge has one endpoint in X and one endpoint in Y. An r*-regular bipartite graph* is a bipartite graph in which every vertex has degree r.

An *independent set* in G is a set of pairwise nonadjacent vertices. The *independence number* of G, written $\alpha(G)$, is the maximum cardinality of an independent set in G. A *matching* in G is a set of pairwise nonincident edges. The *matching number* of G, written $\mu(G)$, is the cardinality of a maximum matching in G. A *vertex cover* in G is a set of vertices $X \subseteq V(G)$ such that every edge of G has at least one endpoint in X. The *vertex cover number* of G, written $\beta(G)$, is the minimum cardinality of a vertex cover in G.

A set $D \subseteq V(G)$ is a *dominating set* if every vertex in $V(G) \setminus D$ has a neighbor in D. The *domination number* of G, written $\gamma(G)$, is the minimum cardinality of a dominating set in G. A set $D \subseteq V(G)$ is a 2-*dominating set* if every vertex in $V(G) \setminus D$ has at least two neighbors in D. The 2-*domination number* of G, written $\gamma_2(G)$, is the minimum cardinality of a 2-dominating set in G. A set $D \subseteq V(G)$ is a *total dominating set* if every vertex of G (including vertices in D) has a neighbor in D; in particular, G must have no isolated vertices. The *total domination number* of G, written $\gamma_t(G)$, is the minimum cardinality of a total dominating set in G.

Fix a graph G. In the *zero forcing process*, each vertex is colored either blue or white. Starting from an initial blue set $B \subseteq V(G)$, we repeatedly apply the color-change rule: If a blue vertex u has exactly one white neighbor v, then u *forces* v, and v is recolored blue. If iterating this rule eventually colors every vertex blue, then B is a *zero forcing set* of G. If in addition, B induces a isolate-free subgraph, then B is a *total zero forcing set* of G. The *zero forcing number* of G, written $Z(G)$, is the minimum cardinality of a zero forcing set in G. The *total zero forcing number* of G, written $Z_t(G)$, is the minimum cardinality of a total zero forcing set in G.

B Integer invariants, predicates, and conventions

Throughout Section 6.1 we work with integers $n \in \mathbb{Z}$ with $n \geq 2$. Every integer $n \geq 2$ admits a unique factorization

$$n \;=\; \prod_{i=1}^{t} p_i^{a_i},$$

where $p_1, \ldots, p_t$ are distinct primes and $a_1, \ldots, a_t$ are positive integers. $\varphi(n)$ denotes *Euler's totient function*: the number of integers $1 \leq m \leq n$ that are relatively prime to n. It is multiplicative and satisfies

$$\varphi(n) \;=\; n \prod_{p \mid n} \Big(1 - \frac{1}{p}\Big),$$

where the product ranges over the distinct prime divisors of n. $\lambda(n)$ denotes the *Carmichael function*: the exponent of the multiplicative group $(\mathbb{Z}/n\mathbb{Z})^{\times}$. Equivalently, $\lambda(n)$ is the least positive integer λ such that $a^{\lambda} \equiv 1 \pmod{n}$ for every a coprime to n. $\mathrm{rad}(n)$ denotes the *radical* of n:

$$\mathrm{rad}(n) \;=\; \prod_{p \mid n} p.$$

Equivalently, if $n = \prod_{i=1}^{t} p_i^{a_i}$ then $\mathrm{rad}(n) = \prod_{i=1}^{t} p_i$. $\omega(n)$ denotes the number of *distinct* prime factors of n; if $n = \prod_{i=1}^{t} p_i^{a_i}$ then $\omega(n) = t$. $\Omega(n)$ denotes the total number of prime factors counted with multiplicity:

$$\Omega(n) \;=\; \sum_{i=1}^{t} a_i.$$

$\tau(n)$ (also denoted $d(n)$) denotes the number of positive divisors of n; if $n = \prod_{i=1}^{t} p_i^{a_i}$ then

$$\tau(n) \;=\; \prod_{i=1}^{t} (a_i + 1).$$

$\sigma(n)$ denotes the sum of the positive divisors of n; if $n = \prod_{i=1}^{t} p_i^{a_i}$ then

$$\sigma(n) \;=\; \prod_{i=1}^{t} (1 + p_i + p_i^2 + \cdots + p_i^{a_i}).$$

The integer n is *square-free* if and only if no prime square divides n. Equivalently, all exponents in the prime factorization are 1, i.e. $\Omega(n) = \omega(n)$. The integer n is *almost square-free* if and only if $\Omega(n) - \omega(n) = 1$, i.e. n has exactly one "extra" prime factor beyond the squarefree regime (counted with multiplicity). For a fixed integer $y \geq 2$, an integer n is *y-smooth* if all prime divisors of n are at most y.

C Finite-group terms and computational conventions

Throughout this appendix, G denotes a finite group. We record the group-theoretic invariants, Boolean predicates, and computational extraction conventions used in Section 6.2.

The *order* of G is $|G|$. The *center* of G is

$$Z(G) \; = \; \{\, z \in G : zg = gz \text{ for all } g \in G \,\},$$

and we record its order $|Z(G)|$. The *commutator* of $x, y \in G$ is $[x, y] = x^{-1}y^{-1}xy$. The *derived subgroup* (also called the commutator subgroup) is

$$G' \; = \; \langle\, [x, y] : x, y \in G \,\rangle,$$

and we record $|G'|$. The *abelianization* of G is the quotient G/G', and we record $|G/G'|$. The conjugacy class of $g \in G$ is

$$g^G \; = \; \{\, x^{-1}gx : x \in G \,\}.$$

We write $k(G)$ for the number of conjugacy classes of G. Equivalently, $k(G)$ is the number of isomorphism classes of irreducible complex representations of G (i.e. $k(G) = |\mathrm{Irr}(G)|$). The *exponent* of G is

$$\exp(G) \; = \; \mathrm{lcm}\{\, |g| : g \in G \,\},$$

where $|g|$ denotes the order of the element $g \in G$.

The finite group G is *abelian* if $xy = yx$ for all $x, y \in G$; G is *cyclic* if there exists $g \in G$ with $G = \langle g \rangle$. A finite group G is *nilpotent* if it is the direct product of its Sylow subgroups. Equivalently, the upper central series terminates at G. A finite group G is *solvable* if its derived series

$$G^{(0)} = G, \qquad G^{(i+1)} = (G^{(i)})'$$

terminates at the trivial group after finitely many steps. G is a *p-group* if $|G| = p^a$ for some prime p and integer $a \geq 0$. (Equivalently, every element has order a power of p.)

For an order cutoff N, the snapshot universe from `SmallGroup` in Section 6.2 is

$$\mathcal{O} \; = \; \big\{\, \texttt{SmallGroup}(n, k) : 1 \leq n \leq N, \; 1 \leq k \leq \#\texttt{SmallGroup}(n) \,\big\},$$

where $\#\texttt{SmallGroup}(n)$ denotes the number of isomorphism types available in the `SmallGroup` library at order n. All invariant and predicate columns in the snapshot table T were computed deterministically in `GAP`. The following commands implement the recorded columns for a group object `G`:

Quantity	GAP command
$\lvert G\rvert$	`Size(G)`
$Z(G)$, $\lvert Z(G)\rvert$	`Centre(G)`, `Size(Centre(G))`
G', $\lvert G'\rvert$	`DerivedSubgroup(G)`, `Size(DerivedSubgroup(G))`
G/G', $\lvert G/G'\rvert$	`FactorGroup(G,DerivedSubgroup(G))`, `Size(...)`
$k(G)$	`NrConjugacyClasses(G)`
$\exp(G)$	`Exponent(G)`
`is_abelian`	`IsAbelian(G)`
`is_cyclic`	`IsCyclic(G)`
`is_nilpotent`	`IsNilpotentGroup(G)`
`is_solvable`	`IsSolvableGroup(G)`
`is_pgroup`	`IsPGroup(G)`

If alternative command names are used in a given `GAP` version or package configuration, we record the exact script used to generate T in the project repository accompanying this paper.

We also list conjectures from Graffiti^3 that the LLM flagged as interesting, or not easily falsifiable.

Conjecture 19 (Graffiti^3). *If G is a finite group with*

$$k(G) \;\geq\; \frac{1}{4}\,|Z(G)| \;+\; \frac{3}{4}\,\sqrt{|Z(G)|\,|G|},$$

then G is solvable.

Conjecture 20 (Graffiti^3). *If G a finite nilpotent group, then*

$$k(G) \;\geq\; \frac{1}{4}\,|Z(G)| \;+\; \frac{3}{4}\,\sqrt{|Z(G)|\,|G|}.$$

Conjecture 21 (Graffiti^3). *If G a finite nilpotent group, then*

$$k(G) \;\geq\; \big|\,|G'| \;-\; |G/G'|\,\big| \;+\; 1.$$

Conjecture 22 (Graffiti^3). *If G is a finite p-group, then*

$$k(G) \;\geq\; \big|\exp(G) \;-\; |G'|\,\big| \;+\; 1.$$

Conjecture 23 (Graffiti^3). *If G is a finite p-group, then*

$$k(G) \;\leq\; \frac{1}{2}\,|G/G'| \;+\; \frac{1}{4}\,|G| \;+\; \frac{1}{2}\,|Z(G)|.$$

Conjecture 24 (Graffiti^3). *If G is a finite p-group, then*

$$k(G) \;\leq\; \frac{1}{2}\,|G/G'| \;+\; \frac{1}{8}\,\mathrm{dl}(G) \;+\; \frac{1}{4}\,|G| \;+\; \frac{1}{2}\,|Z(G)| \;-\; \frac{1}{4},$$

where $\mathrm{dl}(G)$ denotes the derived length of G.

Conjecture 25 (Graffiti^3). *If G is a finite group, then*

$$k(G) \;\geq\; |G/G'| \;+\; \mathrm{dl}(G) \;-\; 1,$$

where $\mathrm{dl}(G)$ is the derived length if G is solvable (and otherwise is taken to be -1).

Conjecture 26 (GRAFFITI[3]). *If G is a finite group, then*

$$k(G) \ \geq \ |G/G'| \ + \ \log_2(\mathrm{dl}(G)),$$

where $\mathrm{dl}(G)$ *is the derived length if G is solvable (and otherwise is taken to be -1).*

Conjecture 27 (GRAFFITI[3]). *If G is a finite p-group, then*

$$k(G) \ \geq \ \frac{7}{16}\,\big|\,|G/G'| \ - \ |\mathrm{Syl}_2(G)|\,\big| \ - \ 4,$$

where $|\mathrm{Syl}_2(G)|$ denotes the order of a Sylow 2-subgroup of G (and is 0 if $2 \nmid |G|$).

Conjecture 28 (GRAFFITI[3]). *If G is a finite group and $G' = [G, G]$ is its derived subgroup, then*

$$\frac{k(G)}{|G|} \ \leq \ |G'|^{-2/5}.$$

D Knot-theoretic terms and computational conventions

Throughout this appendix we record the invariants, predicates, and dataset conventions used in Section 6.3 (and in the associated snapshot table).

Let $K \subset S^3$ denote a knot. $c(K)$ denotes the crossing number of K, i.e. the minimal number of crossings among all diagrams representing K. $\det(K)$ denotes the determinant of K, defined by

$$\det(K) \ = \ |\Delta_K(-1)|,$$

where $\Delta_K(t)$ is the Alexander polynomial normalized in any standard way (since evaluation at $t = -1$ is invariant up to sign). $\sigma(K)$ denotes the (classical) knot signature, i.e., the signature of a symmetrized Seifert form (equivalently the Levine–Tristram signature at $\omega = -1$ under standard conventions). This is the signature used in the obstruction $u(K) \geq \frac{1}{2}|\sigma(K)|$. $v_2(K)$ and $v_3(K)$ denote Vassiliev (finite-type) invariants of orders 2 and 3, respectively, under the normalization used by the data source employed to construct the snapshot table. (Since several normalizations occur in the literature, we treat v_2, v_3 as named columns whose meaning is fixed by the extraction pipeline.) When K is hyperbolic, $Vol(K)$ denotes the volume of the complete hyperbolic structure on $S^3 \setminus K$. In the snapshot table, $Vol(K)$ is recorded only when the geometric pipeline returns a numeric value. $u(K)$ denotes the unknotting number: the minimum number of crossing changes required to transform K into the unknot. `is_hyperbolic`(K) and `fibered`(K) are Boolean predicates taken from the knot census metadata or computed by the chosen software pipeline. Their meaning is fixed by the extraction script used to populate the snapshot table.

We also record one additional conjecture not mentioned in Section 6.3:

Conjecture 29 (GRAFFITI[3]). *If K is a hyperbolic and reversible knot, then*

$$u(K) \ \leq \ \left\lfloor \frac{1}{5}\, c(K) \right\rfloor.$$

E Calabi–Yau terms, arithmetic features, and dataset conventions

This appendix fixes notation and conventions for the Calabi–Yau hypersurface experiment in Section 6.4. Throughout, a data row corresponds to a weighted-projective Calabi–Yau threefold hypersurface $X_{\mathbf{w}} \subset \mathbb{P}^4(\mathbf{w})$ specified by a weight vector $\mathbf{w} = (w_1, \ldots, w_5)$.

For a positive integer weight vector $\mathbf{w} = (w_1, \ldots, w_5)$, the weighted projective space $\mathbb{P}^4(\mathbf{w})$ is the projective orbifold obtained from $\mathbb{C}^5 \setminus \{0\}$ by the $\mathbb{C}^\times$-action

$$\lambda \cdot (x_1, \ldots, x_5) \; = \; (\lambda^{w_1} x_1, \ldots, \lambda^{w_5} x_5).$$

We treat $\mathbf{w}$ as the primary discrete descriptor of the ambient space. We define the (ambient) anticanonical degree

$$d \; := \; \sum_{i=1}^{5} w_i.$$

In the standard weighted-hypersurface construction, a Calabi–Yau hypersurface is realized as a (quasi-smooth) hypersurface of weighted degree d in $\mathbb{P}^4(\mathbf{w})$. In the snapshot table we sort weights nondecreasingly:

$$w_1 \le w_2 \le w_3 \le w_4 \le w_5.$$

All formulas referring to w_3 (the median weight) use this ordering. Each row records Hodge numbers $(h^{1,1}, h^{2,1})$ for the Calabi–Yau threefold $X_{\mathbf{w}}$. These values are taken from the chosen census source (or from a deterministic computation pipeline associated to that source); they are treated as input data rather than recomputed symbolically from scratch in our workflow. We record the Euler characteristic in the convention

$$\chi(X_{\mathbf{w}}) \; := \; 2(h^{1,1} - h^{2,1}),$$

and abbreviate $\chi := \chi(X_{\mathbf{w}})$ when the hypersurface is clear.

In addition to the numeric columns $(w_1, \ldots, w_5)$ and d, we record Boolean predicates that depend only on the weight vector $\mathbf{w}$ and are intended as low-complexity regime markers.

`SD` (sum divisible). $\texttt{SD}(\mathbf{w})$ is `True` if and only if each weight divides the anticanonical degree:

$$\texttt{SD}(\mathbf{w}) = \texttt{True} \quad \Longleftrightarrow \quad d \equiv 0 \pmod{w_i} \;\; \text{for all } i = 1, \ldots, 5.$$

`EO` (exactly one even). $\texttt{EO}(\mathbf{w})$ is `True` if and only if exactly one weight is even:

$$\texttt{EO}(\mathbf{w}) = \texttt{True} \quad \Longleftrightarrow \quad \big|\{\, i : 2 \mid w_i \,\}\big| = 1.$$

`REP` (repeat). $\texttt{REP}(\mathbf{w})$ is `True` if and only if the multiset of weights contains a repetition:

$$\texttt{REP}(\mathbf{w}) = \texttt{True} \quad \Longleftrightarrow \quad \exists\, i < j \;\; \text{with} \;\; w_i = w_j.$$

`PC` (pairwise coprime). $\texttt{PC}(\mathbf{w})$ is `True` if and only if the weights are pairwise coprime:

$$\texttt{PC}(\mathbf{w}) = \texttt{True} \quad \Longleftrightarrow \quad \gcd(w_i, w_j) = 1 \;\; \text{for all } i < j.$$

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